RAD-SMART: Research Report and Implementation Plan

Radiotherapy AI-assisted Dynamic Scheduling, Machine Allocation and Resource Tracking

Health-a-thon 2026 · Cancer track · Doctor / care-team facing use case 04: Clinic Operations & Patient Flow

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> **Assistive, not diagnostic.** RAD-SMART plans *when* and *on which machine* a session takes place. It never decides whether, how or how much a patient is treated. Urgency, technique and machine eligibility are always entered by clinicians and physicists, every plan is a proposal until a person approves it, and every action is logged. The prototype's patients are synthetic. Today's waiting times come from the department's anonymised records (October–December 2024) and are reported only as aggregates.

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1. Executive summary

The problem. A radiotherapy machine treats 70–90 patients a day, most of them every working day for weeks. In our department one Versa HD runs from 08:30 until about 01:00 in three RTT shifts, with a ceiling of 90 patients a day, and a second machine arrives in early 2027. Radiation therapy technologists (RTTs) give out appointment times by hand, without a measure of how many machine-minutes each patient actually needs. Too many patients are told to come at the same time, so they wait, every day of a 5–7 week course. The department's own records (55 days, October–December 2024, 3,137 patient-days) show how: the median wait from arrival to treatment is 34 minutes, but one patient in ten waits more than 109 minutes, 22% are treated more than an hour after their appointment and only 29% within 15 minutes of it. Delays build through the day, from about 19 minutes for patients arriving before 10:00 to about 66 minutes after 19:00, and evening patients now come about 32 minutes early to protect their place. The department must also finish new starts by 17:00, keep complex procedures inside 10:00–17:00 while senior staff are present, keep 13:30–14:00 for blood irradiation, share one breast board and one ABC unit with the CT simulator, treat MHRC patients in time for the 17:00 hospital bus, get public-transport patients home by 21:00, fit same-day urgent starts before 18:00, run total body irradiation (TBI) courses, and recover from machine faults. As the problem statement says, this is not appointment booking. It is dynamic allocation of machine time, staff and shared resources under many constraints.

Our answer. RAD-SMART is an operational layer between the patient list and the machines, built as a hybrid AI system with four parts:

1. **Predict.** A machine-learning model predicts each session's machine-minutes: the expected value used for planning and a cautious value (P80) used to flag fragile plans. The department records arrival and treatment-start times but not how long each session takes, so RAD-SMART starts from the department's own estimate table and learns once room entry and exit are captured.
2. **Optimise.** A constraint-optimisation engine turns those minutes into a minute-level plan for each machine. Hard rules cannot be broken, because the solver will not produce a plan that breaks them, and an independent checker re-verifies every plan. These rules include new starts by 17:00, complex cases in 10:00–17:00, the blood slot, MHRC slots and the 17:00 bus, public transport home by 21:00, and one breast board and one ABC unit shared with the CT simulator.
3. **Simulate.** A digital twin of the department stress-tests every plan against late arrivals, overruns, urgent starts, imaging holds and machine faults. The same engine gives a three-week machine-load forecast that plans new starts around TBI courses, and advises two days ahead how many new patients to start on each machine.

4. **Explain and converse.** A copilot on Sarvam AI's Indian-language models explains every placement in plain language and turns rule changes typed in plain language into versioned, validated configuration. It sends patients and caregivers their reporting time and live updates by WhatsApp, SMS or voice call, in Kannada, Tulu, Malayalam, Hindi or English. The copilot never makes a scheduling decision.

What changed for the build sprint. On 24 October 2026 the department told us how its day really runs, and we built it in:

- **Roles.** The doctor decides the treatment plan and the machine and approves each course. The technologist records arrival, room entry and exit, machine status and any issue.
- **Standing slots.** A patient keeps one time slot until the course ends, unless they ask to change it. The new start is placed before 17:00; every later fraction takes a standing slot, mostly in the evening.
- **Breakdowns of 15 minutes to 48 hours.** Up to 60 minutes, nobody is sent home and everyone is treated with overtime. Over 60 minutes, as many patients are sent back as the expected downtime needs. New starts wait until the machine works, patients who have not left home are told to postpone travel, and a technologist or doctor approves.
- **A working application in CARE.** RAD-SMART now runs as a plugin of CARE, the Open Healthcare Network's open-source hospital information system. It has the intake form with a tick box for every rule, the proposal and approval, the technologist's day board and the breakdown response (Section 6.10). A **RADSMART** button sits on the patient's visit page.

The evidence so far. We built a working proof of concept (PoC) in Python. Its patients are synthetic, its rules are the department's (answers of 21 September and 24 October 2026), and its digital twin is calibrated on the department's records: how appointment times are spread over the day and how early patients arrive at each hour. We test it three ways:

- **the department's own 55 recorded days**, re-planned in the twin;
- **12 weeks of synthetic courses** with standing slots;
- **an 87-patient day** on the Versa HD from 08:30 to 01:00, 10 more patients than the busiest recorded day, loaded to 92.7% and simulated 300 times with realistic randomness.

A separate four-page impact analysis (`docs/RAD-SMART_Impact_Analysis`) compares each dataset with and without RAD-SMART.

What we measured	Without RAD-SMART	With RAD-SMART
The department's 55 recorded days: 90th-percentile wait	108.6 min (recorded)	**70.4 min (−35%)**, re-planned in the twin
The department's 55 recorded days: treated more than 1 hour late	22.2%	**5.7%**
12 weeks with standing slots (about 63 patients a day): 90th-percentile wait	81.6 min	**49.3 min (−40%)**

The 87-patient day, against the department's busiest recorded days (70–77 patients):

What we measured	Today (department records, 70–77-patient days)	RAD-SMART (simulated, 87 patients)
Median wait from arrival to entering the treatment room	38 min	36 min
90th-percentile wait (one patient in ten waits longer)	130 min	**76 min (−42%)**
Treated more than 1 hour after the appointment	27%	**6%**

What we measured	Today (department records, 70–77-patient days)	RAD-SMART (simulated, 87 patients)
Treated within ±15 min of the appointment	30%	**53%**
Median wait, arriving before 10:00 / after 19:00	19 / 66 min (all recorded days)	36 / 36 min
Hard rules in the evening plan (independent check)	not checked	**all 8 met**
45-minute fault at 11:00: median wait without → with live re-planning	–	53 → **37 min**, re-planned in 10 s; nobody sent home; 1 new start moved to the next day
150-minute fault at 10:00: median wait without → with live re-planning	–	104 → **39 min**; 1 patient asked not to travel and sent back; 2 new starts moved to the next day
Two machines and a TBI week: peak planned load	154% on the Versa HD (equal head-count, simulated)	**98% and 99%**

Five findings matter most for a real rollout.

1. **Fewer long waits and kept times, not a shorter typical wait.** The median stays about where it is on today's busiest days, because every patient is asked to report 20 minutes early (45 for pelvic patients). What changes is the tail and punctuality: the 90th-percentile wait falls from 130 to 76 minutes, the share treated more than an hour late from 27% to 6%, and waits no longer build up in the evening: morning patients wait a little longer than today, because they allow 20 minutes to prepare, and evening patients far less. If patients keep arriving as early as they do today, the 90th percentile is 110 minutes, but lateness still falls: 5% are treated more than an hour late.
2. **Benefits from day one.** The optimiser using only average durations already gets the 90th-percentile wait to 82 minutes, so the department benefits before any durations are recorded. Machine learning then adds punctuality (48.9% → 52.9% within ±15 minutes).
3. **The department chooses when the day ends.** Holding 40 minutes for urgent starts gives the shortest waits, with the last patient finishing at 01:03. Holding nothing ends the day at 00:49, with a 90th-percentile wait of 88 minutes.
4. **Rules and recovery.** Every hard rule held in every plan, and both downtime re-plans follow the department's own rules.
5. **Standing slots have a price.** Over 12 weeks the last patient finishes at about 00:07 instead of 21:55, still inside the department's hours. One new patient in five starts later, by 0.4 working days on average, and a patient who asks for a particular time gets it less often, because a standing slot must be free on every day of the course (Section 7.8).

The pilot. 90 days on the Versa HD:

- two weeks adding room entry and exit taps to the arrival and treatment times the department already records, to refresh the baseline;
- three weeks in shadow mode;
- eight weeks live, with the doctor approving every course and its slots, and a technologist or doctor approving every breakdown response.

The primary KPI is the 90th-percentile wait from arrival to linac entry, the long waits patients remember, with a target reduction of at least 25% (from 109 minutes in the records). Punctuality is the main secondary KPI: patients treated more than an hour after their appointment, from 22% to 10% or fewer. It needs about ₹2–3 lakh of hardware and messaging costs and works with the Excel lists, paper registers and WhatsApp the department already uses.

Why this can win. It solves a daily, high-frequency pain for thousands of patients and their caregivers. It uses AI where each technique is strongest: machine learning to predict, mathematical optimisation to guarantee, simulation to test, and a language model to explain and talk to people. It stays inside the hackathon's guardrails by design, it is built on the department's own rules and calibrated on its own records, and it already produces results rather than a mock-up.

2. The problem and why it matters

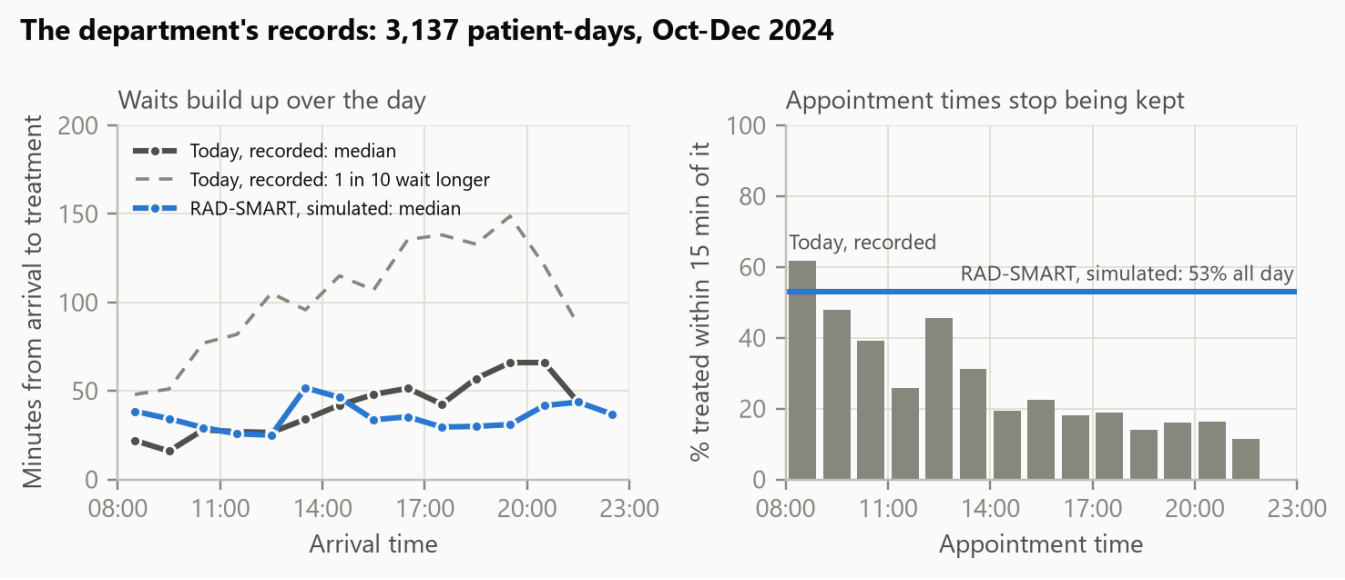
2.1 What happens today

Radiotherapy differs from a normal outpatient clinic. A patient attends every working day for 1–2 weeks (palliative) or 5–7 weeks (curative), so one machine carries 70–90 patients a day. It has to absorb new starts, urgent palliative patients who must start the same day as their consultation, and complex procedures such as SBRT, SRT, TBI and CSI. A routine session occupies roughly 7–15 minutes of machine time, while a new start or complex procedure takes 20–60 minutes.

Appointment times are given by hand. They are not based on how long each treatment takes or how loaded the machine is. As a result, too many patients are asked to report in the same period and other periods sit under-used. Patients have an appointment yet still face long, unpredictable waits.

Our department. One Elekta Versa HD runs from 08:30 until about 01:00–02:00 in three RTT shifts, with an operational ceiling of 90 patients a day. A second machine arrives in early 2027, and its capability matrix is still being agreed. Most patients speak Kannada or Tulu; a few speak English, Hindi or Malayalam. The department records appointment, arrival and treatment-start times, but not how long sessions take.

What the department's records show. The department shared anonymised arrival and treatment records for 55 days, 15 October to 31 December 2024: 3,137 patient-days for 263 patients. The system writes 12:00 AM when a time is missing, so we treat those entries as missing. We fix 3 obvious AM/PM slips and drop 36 impossible entries (a treatment recorded before arrival), which leaves 2,685 waits. We use only aggregates.



The department's records. Left: waits build up through the day (the blue line is RAD-SMART's simulated median, for comparison). Right: the share treated within 15 minutes of their appointment falls from the morning to the evening.

- **Waits.** The median wait from arrival to treatment is 34 minutes and the mean 47 minutes. One patient in ten waits more than 109 minutes; 30% wait over an hour and 7% over two hours. For comparison, Munshi et al. report a mean of 37.4 minutes at a two-linac centre with barcode check-in [3].
- **Delays build through the day.** The median wait is about 19 minutes for patients arriving before 10:00 and about 66 minutes for those arriving between 19:00 and 21:00.
- **Appointment times stop being kept.** Overall 29% of patients are treated within 15 minutes of their appointment: 62% of 08:00 appointments but only 12–16% in the evening. 22% are treated more than an hour late.
- **Patients adapt by coming early.** They arrive a median of 19 minutes before their appointment: about 10 minutes for a morning appointment and about 32 for an evening one. Part of the evening wait is this defensive

early arrival.

- **Volume is rising, and busy days are worse.** Weekdays averaged 54 patients in October and 67 in December; the busiest day had 77. On days with 70–77 patients the 90th-percentile wait is 130 minutes and 27% are treated more than an hour late. The last patient is usually treated around 22:38.

2.2 Constraints the plan must respect

The problem statement lists the rules, and the department confirmed and extended them on 21 September and 24 October 2026 (Appendix B). We classify each as hard (never broken) or soft (traded off), and every one of them is editable configuration, not code.

Rule (department's answer)	Type	How RAD-SMART treats it
Operating day 08:30 to about 01:00–02:00, three RTT shifts, ceiling of 90 patients a day	Hard	Regular day to 01:00; overtime penalised; hard stop at 02:00
Complex treatments (SBRT, SRT, TBI, CSI) need senior staff, 10:00–17:00	Hard	Session starts and ends inside the window, with a 25-min safety buffer
Protected complex slot 12:00–13:30	Soft, released if unused	Complex cases are pulled into it; other patients use it when no complex case needs it
New starts completed by 17:00	Hard	Planned to finish at least 25 min before the cut-off (buffer configurable); 150 min before 17:00 kept free for new starts
A patient keeps one time slot until the course ends, unless they ask to change it; later fractions mostly take evening slots	Hard	A standing slot, free on every treatment date, chosen when the doctor approves the course; a change request is proposed and approved (Section 6.4)
Urgent same-day starts: 0–3 a day, arriving in the afternoon	Hard: treated before 18:00	Two 20-min holds (14:40 and 17:00) that ward and dormitory patients can use on standby; used for catch-up if nobody needs them
Blood irradiation, every day on the Versa HD, 13:30–14:00	Hard	Machine blocked
One breast board and one ABC unit, shared with the CT simulator (open 11:00–18:00); transfer under 2 min	Hard	Counted per unit, with CT-simulator bookings, the 2-min transfer and the linac's reservations (Section 6.7)
Paying patients get their preferred time as far as possible	Hard for the department; modelled as a strong preference	Four times the normal weight, 15-min tolerance; the plan reports how many got their time
MHRC patients: protected 16:00–17:00 slots; must finish for the 17:00 hospital bus	Hard; slots released if unused	Treated between 16:00 and 16:45 after the bus arrives at 15:45
Public-transport patients finish before 21:00	Hard	Latest end 21:00
Older patients: earlier or convenient slots	Soft	Cost for every minute after 13:00
Inpatients, dormitory residents and patients living nearby are flexible	Soft	They take the late-evening slots; when the machine runs ahead, ward and dormitory patients are called in and nearby patients get a "come early" message
Patients report at least 20 min before their slot; pelvic patients (cervix, endometrium, rectum, anal canal, bladder) drink 500 mL of water, wait 30 min and report 45 min before	Instruction to patients	Every message gives the reporting time and preparation; waits are measured from arrival, so they include it

Rule (department's answer)	Type	How RAD-SMART treats it
Imaging review: the oncologist may decline to treat; the patient is repositioned and treated at once or later the same day	Event; buffer needed	Re-queued automatically; the urgent holds and the late evening absorb re-treatments
TBI: two fractions a day of 60–90 min (first in the morning and in the evening), about every 2–3 months, known a month ahead; new starts tapered from a week before	Hard (planned)	Slots blocked on TBI days; the forecast holds back exactly the new starts that would not fit
Machine capability (Versa HD, second machine from early 2027)	Hard	Eligibility matrix from physics; pending departmental consensus
Machine downtime: 15 min to 48 h; up to 60 min everyone is treated with overtime; over 60 min patients are sent back as the expected downtime needs	Event with rules	The department's downtime rules (Section 6.5)
Staff overtime	Soft	Penalised in the objective; up to 60 min on a breakdown day
Approvals: the doctor approves the plan and machine for each course; a technologist or doctor approves a breakdown response	Human	Nothing reaches a patient before that approval

2.3 Evidence from India and elsewhere

- **Waiting is a daily, measurable burden.** An audit of 34,438 sessions at an advanced Indian centre with two linacs and barcode check-in found a mean gross wait (arrival to linac entry) of 37.4 minutes and a mean total time in the department of 52.4 minutes on routine days. Mean set-up plus treatment time was only 15.1 minutes [3]. Centres without digital queueing report much longer waits. At Tata Memorial Centre, a quality-improvement project cut the median first-day wait from 6 to 4.5 hours [5].
- **Communication is the best-rated fix.** A survey of Indian radiation oncology departments identified two-way communication with patients as the most effective waiting-time strategy [4].
- **Capacity is scarce, so every minute counts.** India has about 823 radiotherapy machines, roughly 0.6 per million people, with a shortfall estimated at 1,209 machines. 69% of the 554 facilities have only one machine [6], which means one bad day on that machine affects every patient in the centre.
- **Repetition multiplies the harm.** A patient waiting an extra hour a day on a 25-fraction course loses about 25 hours, and so does the caregiver who usually comes with them.

2.4 What changes if we solve it

For patients, the plan means predictability: they arrive close to their real treatment time and hear about delays before they leave home. RTTs no longer have to juggle times by hand or reshuffle the queue after every disruption. Clinicians get complex procedures reliably placed while supervision is available. The department gets smoother machine use, less overtime, objective new-start planning and data for future capacity decisions.

3. Fit with Health-a-thon 2026

Hackathon expectation	How RAD-SMART meets it
Non-clinical, operational workflow problem	Scheduling, queueing and resource tracking. It uses operational tags only: technique, fraction number, accessories, logistics flags
Out of scope: diagnosis, treatment recommendations, clinical decision support, risk scoring, interpretation of medical data	It does not read images, doses, diagnoses or notes, and it does not decide urgency or eligibility. Clinicians and physicists enter those

Hackathon expectation	How RAD-SMART meets it
Human override on every automated step	Every course, plan and re-plan is a proposal; the doctor approves courses and slots, a technologist or doctor approves the day and every breakdown response, and the reason for an override is captured
Audit trail	Append-only log of plans, edits, overrides, rule versions, messages and model versions
Measurable KPI with baseline	90th-percentile wait from arrival to linac entry; the baseline is already in the department's records and is refreshed in the pilot's first two weeks; secondary and balancing KPIs in Section 8
Pilot evidence in 60–90 days	90-day plan: baseline, shadow mode, then live, analysed as an interrupted time series
Light integration; works with paper, Excel, WhatsApp	Runs inside CARE as a plugin, with a RADSMART button on the patient's visit page and no change to CARE's code. Imports the Excel list or a photo of the paper register. QR check-in. OIS integration only in phase 2, read-only
Multilingual and caregiver-friendly	Messages in the patient's language by WhatsApp, SMS or voice, with an optional caregiver copy. Spoken "running late" replies are understood
Leverage existing AI capabilities and frameworks	Sarvam AI (LLM, speech, translation, OCR), LangGraph from the LangChain ecosystem for tool orchestration, OR-Tools, scikit-learn
Easy for busy clinics	Evening-before plan in one screen; live board; one-tap approve; printable fallback list

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4. State of the art and the gap

Operations research in radiotherapy is proven, but mostly for choosing the day. A literature review of radiotherapy resource planning shows decades of work on patient scheduling [7]. Dynamic programming and Markov models allocate treatment start days to incoming patients [11]. Prediction-based online policies decide how long each new patient should wait for a start date [12], and column-generation methods handle machine unavailability [22]. In practice, a MILP scheduler implemented at two Dutch centres produced weekly schedules in 5 minutes instead of 1.5 days of manual work and cut the variation in daily appointment times by 51% [8]. Time-window preferences can be honoured for almost all sessions [9]. At a ten-linac Belgian centre, automated scheduling reduced the average wait to start treatment by 80% [10]. These systems mostly optimise *which day*, in large multi-machine, fully digital European centres.

Duration prediction works. Models trained on positioning and treatment times predict session length well enough to schedule by the minute [13], and earlier regression-tree models reached 84% accuracy for treatment duration [14].

New AI methods are being tried, with trade-offs. Deep reinforcement learning has been applied to integrated pre-treatment and treatment scheduling [15]. A 2026 preprint tests large language models directly as radiotherapy schedulers under realistic constraints [16]. Neither gives the verifiable guarantee a department needs for rules such as "no new start after 17:00".

The gap RAD-SMART fills.

1. **Within-day, minute-level sequencing.** Sessions are placed by predicted minutes, with shared accessories and CT-simulator bookings in the same model.
2. **Live recovery.** When the machine is down or a patient is late, the rest of the day is re-planned in seconds while keeping changes small, and the right patients are told in time.
3. **The Indian setting.** It is built for one- or two-machine centres, mixed Elekta and Varian fleets, patients who depend on public transport, low digital literacy, many languages, and paper or Excel workflows.

4. **Safe use of language models.** The LLM is the interface for explanations, rule editing and multilingual messages. It is not the decision-maker, so hard rules are guaranteed by the solver while people still talk to the system in plain language.

As far as we could find, no published or commercial system combines prediction, guaranteed-constraint optimisation, simulation, plain-language explanation and multilingual patient messaging in a light-integration package for this setting.

5. Five approaches considered

We considered five ways to build RAD-SMART and scored them against the hackathon's priorities.

A. Rule-based slot templates. Fixed slot lengths by technique (for example 10 minutes routine, 30 minutes new start), filled first-come-first-served, with block rules. It is simple, transparent and quick to build. But it cannot absorb variation, balance machines by minutes, or recover from disruption, and staff end up overriding it constantly.

B. LLM-only agent. A large language model receives the patient list and the rules in a prompt and writes the schedule. It is flexible, conversational and quick to demo. However, it cannot guarantee hard constraints, is weak at arithmetic packing, gives different answers to the same input, and is hard to audit. That is unacceptable when a missed rule means an unsupervised complex treatment.

C. Classical optimisation, static. A mixed-integer model builds the day plan from average durations each evening. It is proven in the literature and guarantees hard rules. On its own, though, it has no learning, no live recovery, no forecasting, and no way to talk to patients or staff.

D. Deep reinforcement learning. A policy is trained in simulation to decide appointment times. It is adaptive in principle, but it needs large volumes of data and simulation, acts as a black box, offers no guarantee on hard rules, and is hard to reconfigure for another department.

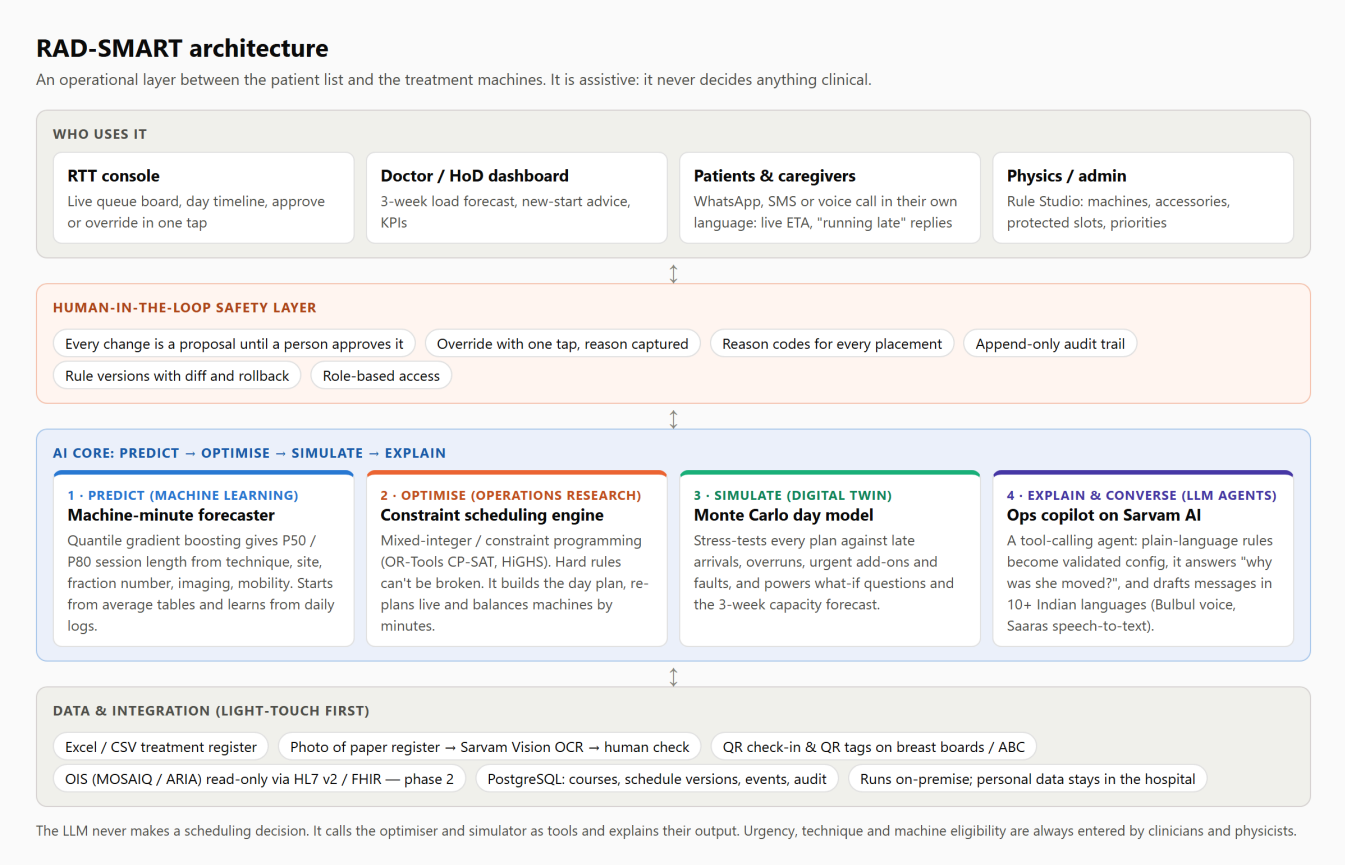
E. Hybrid: predict, optimise, simulate, explain (selected). Machine learning predicts minutes. Constraint optimisation builds and repairs the plan with guaranteed rules. A digital twin tests plans and forecasts load. An LLM copilot explains the plan, edits rules and speaks the patient's language. Each technique does what it is best at, and each can be switched off without breaking the others.

Criterion (weight)	A. Rule templates	B. LLM-only	C. Static optimisation	D. Deep RL	E. Hybrid
Safety and scope compliance (20%)	5	1	5	2	5
Impact on waits and utilisation (20%)	2	2	4	4	5
Real-time adaptivity (15%)	1	3	2	4	5
Feasibility in the sprint and a 90-day pilot (15%)	5	3	4	1	4
Explainability and audit (10%)	4	2	4	1	5
Ease of use and multilingual reach (10%)	2	5	2	2	5
Configurability for other departments (10%)	2	4	3	2	5
Weighted score (out of 5)	**3.10**	**2.60**	**3.60**	**2.45**	**4.85**

Scores run from 1 (poor) to 5 (excellent). The hybrid scores lower only on feasibility, because it has more parts. We handle that by building it in layers. The optimiser with average tables works on its own from week 1, and machine learning, simulation and the copilot are added on top.

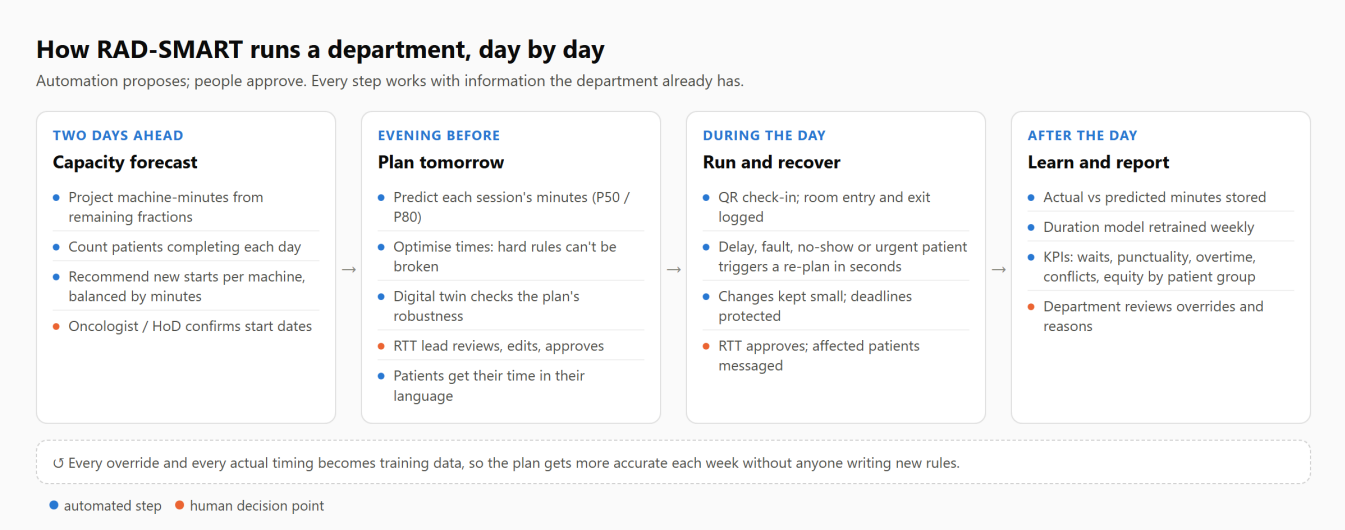
6. The recommended solution

6.1 Architecture



RAD-SMART architecture. Four user groups work through a human-in-the-loop safety layer; the AI core predicts, optimises, simulates and explains; data enters through light-touch channels first.

6.2 How a day runs



The daily loop. Blue steps are automated; orange steps are decisions made by people.

1. **Two working days ahead.** The capacity forecast projects each machine's committed minutes from every patient's remaining fractions, counts completions, and recommends how many new patients to start on each machine. The oncologist or head of department confirms who starts and when.

2. **When a course is approved.** The doctor enters the prescription and ticks the priority checklist. RAD-SMART proposes a new-start slot before 17:00 and one standing slot for every later fraction, free on every treatment date, with its reasons and alternatives. The doctor approves, and the patient keeps that time until the course ends unless they ask to change it.
3. **Evening before.** The day's plan is the standing slots plus the new starts. The independent checker re-verifies every hard rule, the digital twin checks how robust the plan is, and the technologist approves it. Patients receive their reporting time in their own language.
4. **During the day.** The technologist records arrival, room entry and exit, and the machine's status. A delay, fault, no-show, urgent add-on or a patient's "running late" reply triggers a re-plan proposal within seconds. When a technologist or doctor approves it, affected patients who can still act on it are messaged.
5. **After the day.** Actual and predicted minutes are stored. The model is retrained weekly, and a new version goes live only after human approval. The KPI dashboard updates, and the department reviews overrides and their reasons.

6.3 Module 1: Predict machine-minutes

- **Inputs** are operational tags already on the treatment card: technique (palliative, 3D-CRT, IMRT, VMAT, breast, breast with DIBH/ABC, SBRT, SRT, CSI, TBI), site group, fraction number (first fraction or not), imaging (none, kV or CBCT), mobility (walking, wheelchair or stretcher), accessories and machine.
- **Model.** Gradient-boosted regressors give the expected minutes, the median (P50) and a cautious value (P80). The plan reserves the expected minutes, because session times are right-skewed and planning at the median would under-book the day. It can use P80 for long or variable sessions such as new starts and complex cases, and P80 also flags fragile plans.
- **Cold start.** The department records arrival and treatment-start times but not session durations, so RAD-SMART starts from the department's own estimate table (minutes per technique, first fraction and accessory). The PoC shows that the optimiser plus averages already captures most of the benefit (Section 7). Timestamp capture starts on day 1 of the pilot.
- **Learning loop.** Every session's actual time is logged by QR or tablet taps, or from OIS timestamps where available. The model is retrained weekly with a drift report, and a new version goes live only after human approval.

6.4 Module 2: Optimise the day

Formulation. The day is a grid of 2-minute steps. Each session is placed on an eligible machine at a start time. The model is a mixed-integer program, and the production version will use constraint programming with interval variables.

```
Decide   start time and machine for each session (or "unscheduled", heavily penalised)
Subject to (hard):
  - one session at a time per machine; operating day 08:30 to 02:00
  - machine blocks: blood irradiation 13:30-14:00, TBI slots, QA, faults
  - machine eligibility from the physics capability matrix
  - complex sessions inside 10:00-17:00 (senior staff), with a 25-min buffer
  - new starts finish by 17:00 minus a 25-min buffer; urgent starts by 18:00
  - MHRC patients in the 16:00-17:00 slots, finished for the 17:00 bus
  - public-transport patients finish by 21:00
  - accessory units (1 breast board, 1 ABC): linac sessions + CT-simulator
    bookings <= units, with 2-min transfers and the linac's reservations
  - capacity holds for same-day urgent starts (released if unused)
Minimise (soft, per 10 minutes):
    w1 x time away from the usual / preferred time beyond a tolerance
         (paying patients: 4x weight, 15-min tolerance; flexible patients: 0.2x)
  + w2 x older patients after 13:00
  + w3 x non-flexible patients finishing after 21:00 + w4 x anyone after 23:00
  + w5 x MHRC patients waiting after 16:00
  + w6 x complex minutes outside the 12:00-13:30 protected block
  + w7 x overtime after 01:00
  + (live mode) w8 x minutes moved for patients who can still be told
              + w9 x extra waiting of patients already in the department
  + 1000 x unscheduled sessions (x5 for new starts and complex cases)
```

All weights, windows and buffers are configuration that the department can see and change.

Solving a 70–90-patient day in under a minute. A single large model over a whole day is slow. RAD-SMART solves it coarse-to-fine in two stages.

1. Stage 1 levels the load in 15-minute buckets. Complex cases, new starts and MHRC patients, the "big rocks", are placed as whole units, and routine patients are spread continuously.
2. Stage 2a places the big rocks exactly, near their stage-1 bucket, and locks them.
3. Stage 2b places everyone else on the 2-minute grid, within ±50 minutes of their bucket.
4. If a patient cannot be placed, that patient's window is opened fully and the solver tries again.

On a laptop with the open-source HiGHS solver, the machine-learning plan for the PoC's 87-patient day reached a proven optimum in 7.1 s, and the average-table plan in 30.9 s. Neither left a patient unplanned. The model has about 4,800 variables and 1,662 constraints. Timings vary with the laptop's load.

Standing slots: a patient's time stays put. The department keeps each patient's slot until the course ends (24 October 2026), and RAD-SMART now does the same. When the doctor approves a course, the slot allocator chooses two things:

1. **The new-start slot**, on the first day, finishing before 17:00 on an eligible machine, with the new start's longer minutes.
2. **One standing slot for every later fraction**: a time that is free on *every* treatment date of the course (daily, alternate days, twice a day, twice or once a week), keeps every hard rule, and leaves 150 minutes before 17:00 free for future new starts. Among the free times it picks the one with the lowest soft cost: the patient's preferred time, the evening for flexible patients, earlier for older patients. Most later fractions therefore go to the evening.

If no new-start gap is left on the requested date, it proposes the first working day with one. When a patient asks for another time, **Find a time** proposes the nearest time that keeps every rule (and, separately, the nearest one inside the new-start reserve, for the approver to weigh), and the slot changes from the chosen date only when someone approves it. A missed fraction is added at the end of the course, in the same slot. The independent checker adds a ninth rule for these days: every patient is in their standing slot. The day optimiser above is still used for breakdowns, for the single-day tests in Section 7 and for any day the department chooses to re-plan in full.

Reason codes. Every placement comes with machine-readable reasons, such as "new start: finish by 17:00", "MHRC: protected slot 16:00–17:00, back on the 17:00 bus", "public transport: finish by 21:00", "report at 09:41 (45

min before: drink 500 mL of water on arrival)", "accessory check: breast board free (CT-simulator bookings and reservations respected)" or "dormitory: late-evening slot, keeps earlier slots for stricter constraints". These drive the RTT console and ground the copilot's explanations.

6.5 Module 3: Live re-planning

When an event arrives (machine fault, overrun, no-show, urgent add-on, imaging hold, or a patient replying "running late"), RAD-SMART does the following:

- It freezes what has already started and re-solves the rest of the day.
- It costs two kinds of change differently. Every minute of delay for a patient already in the department, or on the way, is a minute in the waiting room, so that waiting is costed heavily, and steeply beyond an hour. A patient still at home can be told a new time, so moving them costs less. The people already waiting are treated first.
- It never asks a patient to come *earlier* than they were told unless they can be reached in time.
- It messages only patients whose appointment is at least 45 minutes away and whose time moves by at least 5 minutes, so it does not flood patients or staff.
- It keeps every hard rule. For example, a new start pushed by a fault still finishes by 17:00.
- It presents the change as a proposal to a technologist or doctor, showing the patients affected, the minutes moved and anyone sent back, and waits for approval.

The department's downtime rules (24 October 2026) are built in. A breakdown lasts from 15 minutes to 48 hours. The technologist enters the expected downtime, or the exact one when the engineer knows it, and RAD-SMART proposes:

Downtime	Department's rule	What RAD-SMART does
Up to 60 min	Everyone is accommodated with overtime	Nobody is sent home. Everyone is treated today, with up to 60 minutes of overtime. Patients already waiting are treated first when the machine is back
Over 60 min	Patients are sent back according to the expected downtime	Sends back only as many patients as are needed to free the machine-minutes that 60 minutes of overtime and today's gaps cannot absorb. Those who have not set out are chosen first, palliative before other radical courses. A gap-sensitive radical course (RCR category 1) or an MHRC patient on the bus is never sent back automatically: the oncologist decides. The missed fraction is added at the end of the course
Any length	New starts are postponed until the machine works	The day's new starts wait for the machine and still finish by 17:00, or move to the next working day
Any length	Patients not yet reported are told to postpone travel	Patients at least 45 minutes from their time are told not to set out until they get a new time; the message is drafted in their language
Any length	A technologist or doctor approves	Nothing reaches a patient before approval. **Re-plan with this estimate** handles an engineer's revised estimate, and **Back online** ends the breakdown

Imaging review holds. After imaging, the oncologist may decline to treat. The patient is then repositioned and treated at once, or treated later the same day. RAD-SMART re-queues the patient automatically, and the urgent holds and the late evening give the buffer the department asked for.

6.6 Module 4: Capacity forecast and new-start advisor

For each machine and each of the next 15 working days, RAD-SMART does the following:

1. It computes the committed load: the predicted minutes of every patient still on treatment, from their remaining fractions.
2. It counts the patients completing each day.

3. It takes the pipeline of patients whose plans will be ready and assigns start days from two working days ahead, each patient to the eligible machine with the most free minutes. It keeps every day of their course within a target utilisation (92%) after reserving capacity for urgent starts.

The output reads like the example in the problem statement, for instance "Wednesday: several patients completing, recommend 5 new starts on Versa HD, 12 on Halcyon". Clinicians decide *which* patients start. RAD-SMART advises *how many* and *where*, balanced by minutes rather than head-count.

TBI courses. Total body irradiation takes two fractions a day of 60–90 minutes, the first of the morning and one in the evening. It comes about every 2–3 months and is known a month ahead. Today the department tapers new starts by hand from a week before. RAD-SMART takes the TBI slots off that machine's capacity on the TBI days and holds back exactly the new starts whose courses would not fit.

The second machine. Its capability matrix is still being agreed by the department. RAD-SMART only ever allocates by the matrix the physicists approve; the PoC uses a placeholder.

6.7 Module 5: Shared resource tracking

The department has one breast board and one ABC unit. Both are shared by the linac and the CT simulator, which runs from 11:00 to 18:00, and moving one between rooms takes under 2 minutes. The department reserves them for the linac at set times:

- both accessories before 11:00;
- both accessories from 16:00 to 17:00, for new starts and MHRC patients;
- the ABC from 12:00 to 13:30;
- the breast board, on the linac only, after 18:00.

A reserved accessory can be released to the CT simulator when the linac does not need it. RAD-SMART models each unit, the CT-simulator bookings, the transfer time and the reservations. Simulator staff enter the bookings or they are imported from the simulator's list, and each one is checked against the reservations. Each accessory carries a QR tag scanned when it leaves or arrives in a room, so the system knows where it is. Conflicts are prevented in the plan. Anything that happens live, such as an accessory not returned, raises an alert before the patient is called.

6.8 Module 6: Copilot and multilingual communication (Sarvam AI)

For staff (RTT console and Rule Studio):

- "*Why was patient P034 moved to 11:25?*" The copilot answers from the placement's reason codes and the re-plan log, and never from guesswork.
- "*From next Monday the blood irradiation slot is 2:00–2:30 on Fridays.*" The copilot converts the sentence into a structured rule, validates it against the schema, shows a before/after diff and a digital-twin impact estimate, and saves it as a new rule version only after an authorised person approves. Earlier versions can be restored with one tap.
- "*What happens if we start 3 extra patients on Wednesday?*" The copilot calls the forecast and simulation tools and summarises the result.

For patients and caregivers:

- **Evening-before message** with the next day's reporting time and preparation, such as "please come at 09:41 and drink 500 mL of water when you arrive" for pelvic patients. It uses the patient's chosen language and channel: WhatsApp, SMS, or an automated voice call for patients who do not read messages.
- **Live updates** when the day moves: "The machine is running about 30 minutes late. Your new time is 11:25. You do not need to come earlier."
- **Voice replies.** A patient can say "the bus is late, I will be 20 minutes late" in their own language. Speech-to-text and intent detection turn it into an event on the RTT console.

- **Caregiver copy.** With the patient's consent, the same message goes to a family member.
- **No clinical content.** Messages carry times and logistics only. Any health question gets the reply "please speak to your doctor or nurse".

Sarvam AI components [17]:

Need	Sarvam component
Copilot reasoning and tool calling	Sarvam-105B (128K context; 10 Indian languages + English)
Voice calls to patients	Bulbul v3 text-to-speech (11 languages, including Kannada, Malayalam, Hindi and English) [23]
Understanding spoken replies	Saaras v3 speech-to-text (the 22 scheduled Indian languages and English, code-mixed speech) [24]
Message translation	Sarvam Translate / Mayura
Paper register import	Sarvam Vision (document OCR, 23 languages)

Tulu. Many of the department's patients speak Tulu, which is usually written in Kannada script. Sarvam's current models do not cover it: Bulbul v3 speaks 11 languages [23], and Saaras v3 transcribes the 22 scheduled Indian languages and English [24]. Tulu is not among either. RAD-SMART therefore sends Tulu speakers the text in Kannada script with a short voice note. Tulu-speaking staff record the note once for each approved template, and the time is filled in from recorded number clips. Patients who prefer it get Kannada speech instead. Spoken replies in Tulu reach the RTT console as audio for a staff member to hear. If Sarvam or another provider adds Tulu, generated speech replaces the recorded clips.

Guardrails for the language model:

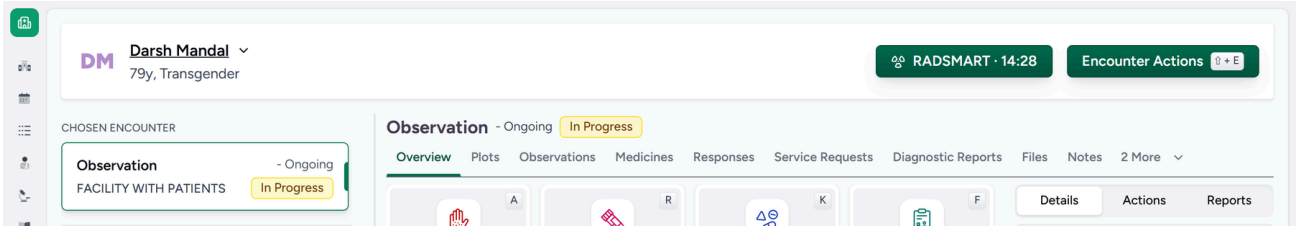
- The LLM runs through an allow-list of tools orchestrated with LangGraph. It can read plans and propose rule changes, but it cannot write a schedule or send a message on its own.
- It sees pseudonymous IDs and operational tags only. Names and phone numbers are added on the hospital's server when a message is sent, so personal identifiers never reach the model.
- Patient messages come from fixed templates. Each template's translation is produced once, checked by a native-speaking staff member and approved, which WhatsApp requires anyway. The model only fills in times.
- If the model or the internet is unavailable, the console shows the raw reason codes and the templates still work, so the scheduling itself never depends on the LLM.

6.9 Safety, override and audit

- Every automated output is a proposal, including plans, re-plans, deferrals, new-start advice, rule changes and new model versions. As the department specified, the doctor approves each course, its machine and its slots, and a technologist or doctor approves the day's plan and every breakdown response.
- An override takes one tap and asks for a reason (a pick-list plus optional free text). Overrides feed weekly review and model learning.
- An independent checker, separate from the solver, re-verifies every hard rule before a plan can be published.
- The append-only audit log records who did what and when, which rule and model versions were used, and every message sent.
- A manual fallback is always available: a printed list, and a one-click switch back to manual mode.

6.10 The application in CARE

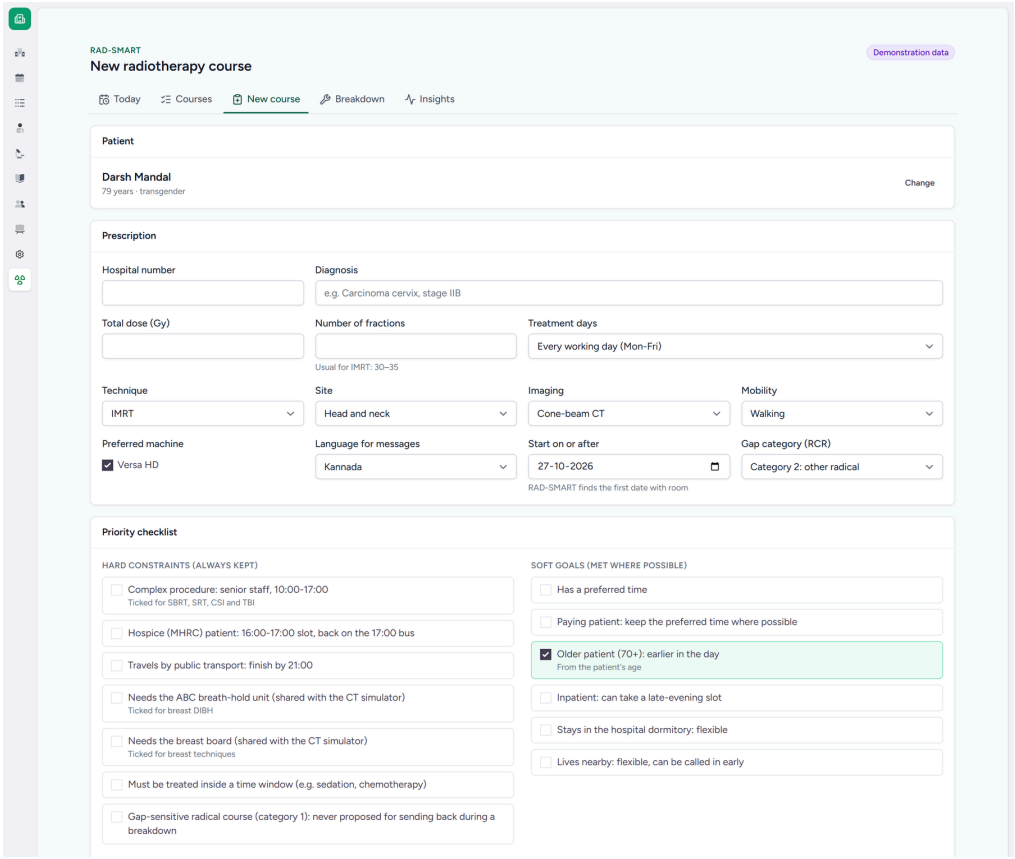
The hackathon asked teams to build on CARE, the Open Healthcare Network's open-source hospital information system. RAD-SMART runs inside it as two plugins (folder `app/`): a Django backend at `/api/care_radsmart/`, which calls the PoC's engine, and a frontend loaded into CARE at runtime. CARE's own code is unchanged. The same pattern, a button on the visit page, applies to the hospital's own EMR through its integration route. Every patient in the screenshots is synthetic.



The patient's visit page in CARE. The RADSMART button opens the patient's course; once a course is approved it shows the standing time.

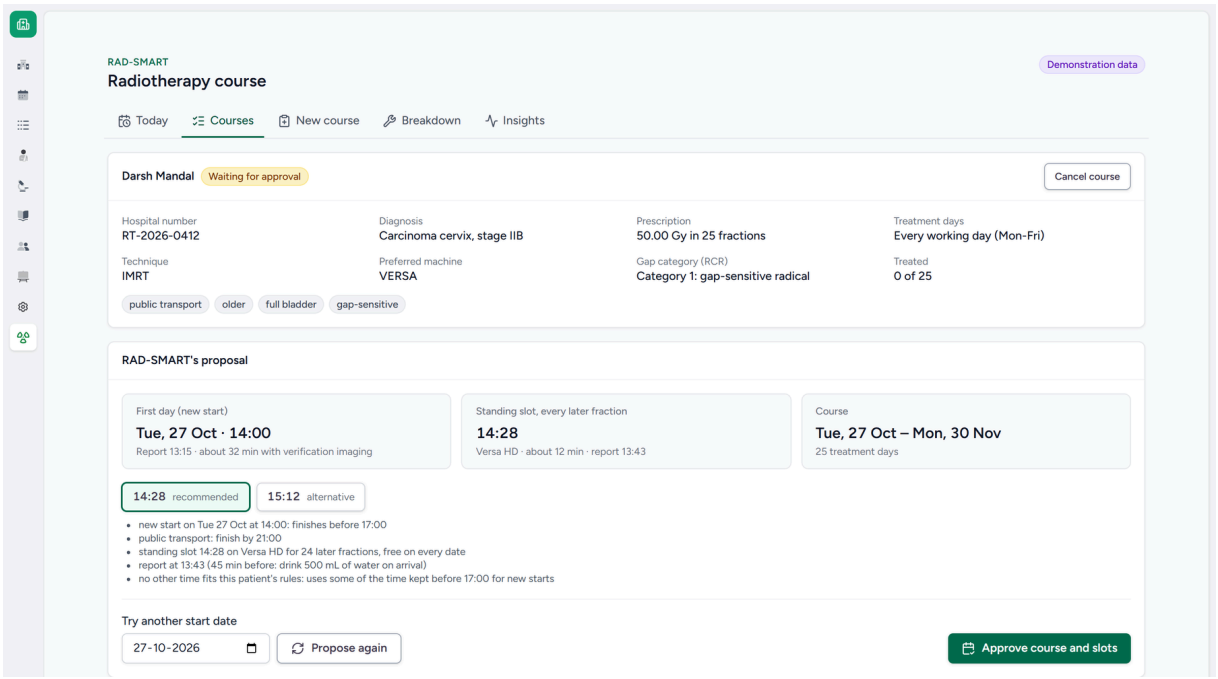
The doctor clicks **RADSMART** on the visit page and fills in the course:

- the hospital number, diagnosis, total dose, number of fractions and treatment days (every working day, alternate days, twice a day, twice a week or once a week);
- technique, site, imaging, mobility, preferred machine, language for messages and the earliest start date;
- the **priority checklist**: a tick box for every hard rule (complex procedure, hospice bus, public transport, ABC unit, breast board, a time window, gap-sensitive course) and every soft goal (preferred time, paying, older, inpatient, dormitory, lives nearby). Age and technique tick their own boxes, and the rules that always apply are listed under the checklist.



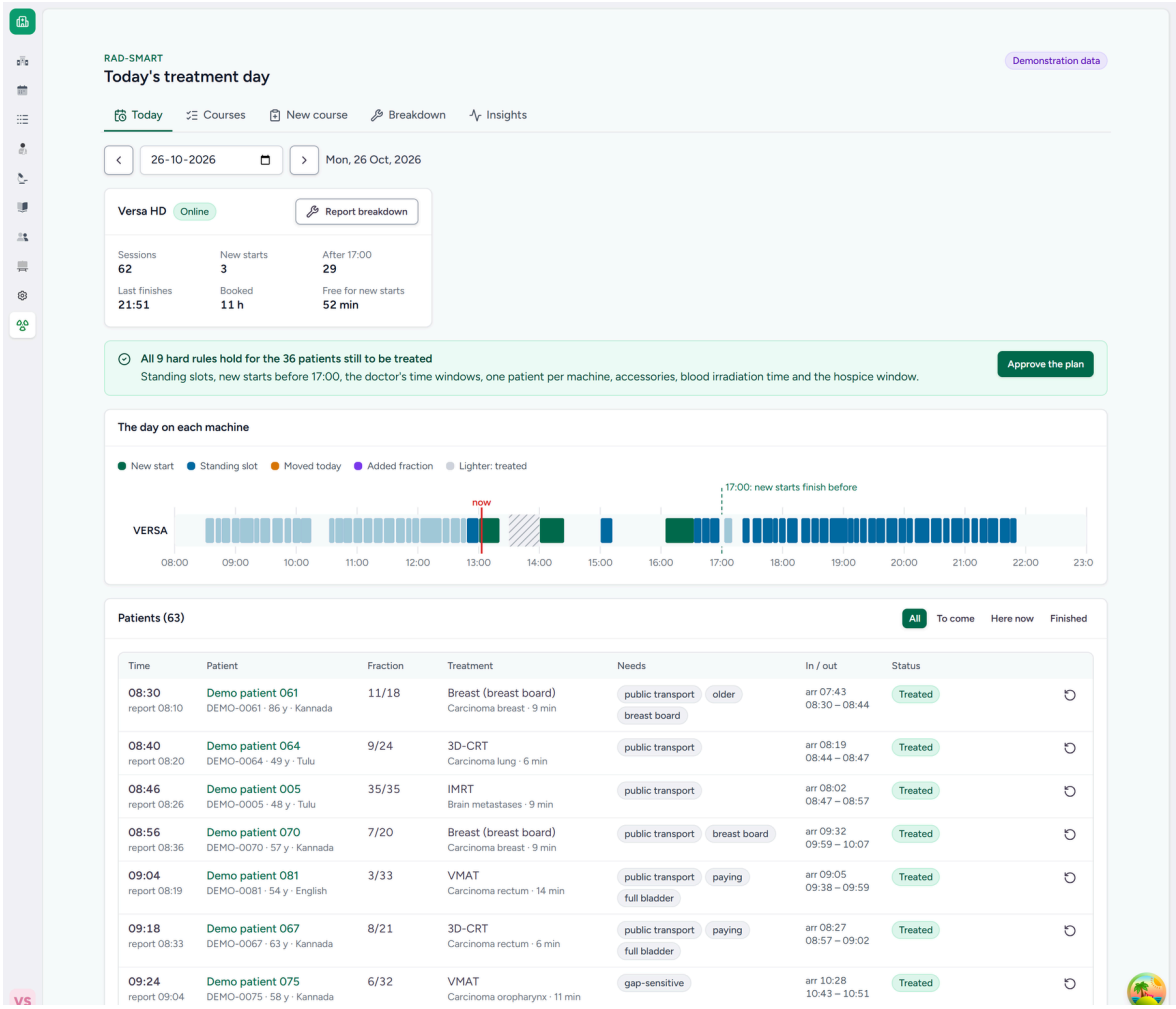
Intake: the prescription and the priority checklist.

RAD-SMART proposes the new-start slot and the standing slot, with its reasons and alternatives. The doctor approves the course, its slots, or asks again from another date.



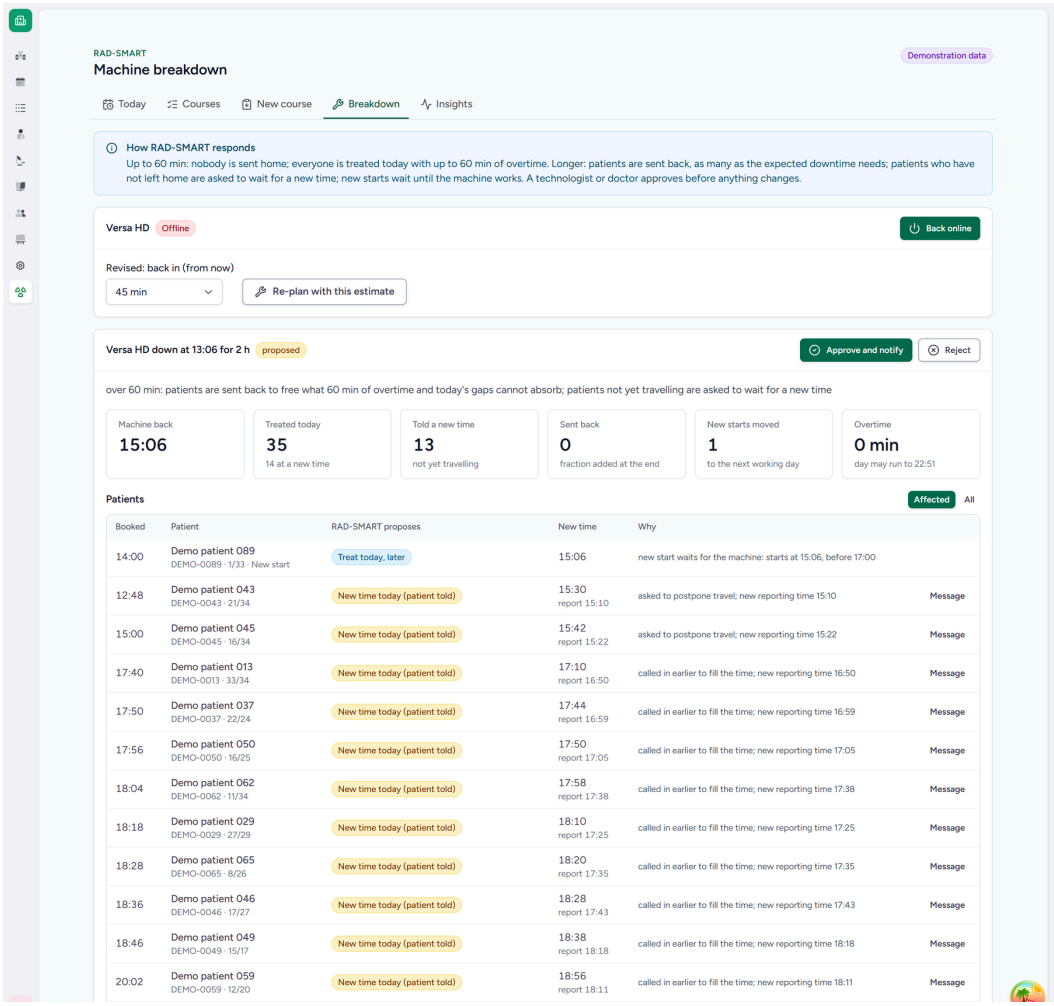
RAD-SMART's proposal for a new course: a new start before 17:00 and one standing slot for the 24 later fractions.

The technologist runs the day from **Today**: each machine's status, the rule check with **Approve the plan**, the day on a timeline (17:00 line, blood-irradiation block, the current time) and the patient list, where they record **Arrived** → **In room** → **Out**. **Not today** adds a missed fraction at the end of the course. Slot-change requests wait on **Courses** for a decision.



Today: the machine, the rule check, the timeline and the patient list with recorded times.

When the machine fails, **Report breakdown** takes it offline with the expected downtime (15 minutes to 2 days). RAD-SMART proposes the response under the department's rules, with every patient's new time, the reason and a message drafted in their language. A technologist or doctor approves or rejects it.



A two-hour breakdown: RAD-SMART's proposal waits for approval. Patients still at home are told to postpone travel; those who can come earlier fill the time.

Insights shows the measured wait from arrival to the treatment room, from the times technologists record, against the baseline (108.6 minutes at the 90th percentile) and the target (25% lower).

Permission	Who has it	Allows
View	doctor, nurse, staff, facility administrator	the pages and read-only data
Operate	doctor, nurse, staff, facility administrator	recording times and machine status, approving the day, breakdown responses and slot changes
Approve courses	doctor, facility administrator	entering, approving and cancelling courses

The permissions are checked on the server; the pages only hide what a role cannot do. Messages are drafted, not sent: sending needs the hospital's SMS or WhatsApp gateway. Setup steps, the API and the limitations are in `app/README.md`.

7. Proof-of-concept evidence

7.1 What we built and how we tested it

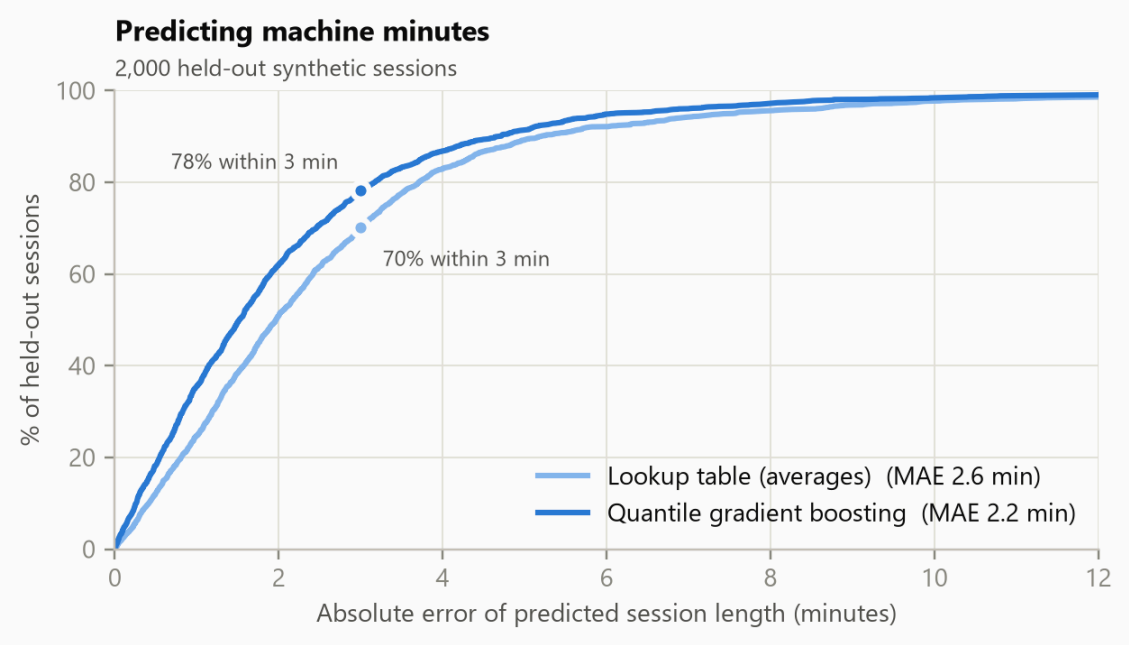
The PoC (folder `poc/`) implements Modules 1, 2, 3 and 4, the standing-slot allocator and the breakdown triage in about 3,700 lines of Python. It uses scikit-learn for the duration models, SciPy's interface to the open-source HiGHS solver for the MILP, and NumPy for the simulator. All patients are synthetic. The department's answers of 21 September and 24 October 2026 are coded as configuration, and its records calibrate the twin. The CARE application (Section 6.10) calls the same engine.

- **Synthetic department, with the department's rules:**
  - operating day 08:30–01:00, with overtime to 02:00;
  - blood irradiation 13:30–14:00;
  - complex cases in 10:00–17:00, with the 12:00–13:30 protected block;
  - new starts finish by 17:00, with a 25-minute buffer;
  - urgent starts by 18:00, with 20-minute holds at 14:40 and 17:00;
  - MHRC patients in the 16:00–17:00 slots, arriving by bus at 15:45 and finished by 16:55;
  - public-transport patients finished by 21:00;
  - one breast board and one ABC unit shared with the CT simulator (11:00–18:00), with the department's reservations, a 2-minute transfer, and five CT-simulator accessory bookings checked against the reservations.
- **Test day.** 87 patients on the Versa HD, near the department's ceiling of 90 and 10 more than the busiest recorded day. They need 890 of 960 available minutes, a load of 92.7%, plus 0–3 urgent starts.
  - Mix: 5 new starts, 3 complex, 3 MHRC and 26 public transport.
  - 34 flexible patients: 5 inpatients, 10 dormitory residents and 19 living nearby.
  - 31 paying (17 with a preferred time), 9 older and 26 pelvic.
  - Accessories: 15 breast board, 5 ABC.
  - Languages: 49 Kannada, 22 Tulu, 9 Malayalam, 5 English, 2 Hindi.
- **Session lengths are illustrative.** The department does not record how long sessions take, so the synthetic techniques' median minutes are our estimates. They will be replaced by the department's estimate table and then by measured times. The records' median gap between treatment starts, 10 minutes, matches the test day's average of 10.2 minutes per patient.
- **Today's booking, calibrated on the records.** Each patient gets an individual appointment on a 5-minute grid, drawn from the department's recorded spread of appointment times on days of similar size, inside the patient's rule window: complex cases and new starts 10:00–16:00, MHRC 16:00, public transport before 19:00, flexible patients in the evening. Treatment is first-come-first-served.
- **RAD-SMART.** Each patient is told a reporting time 20 minutes before the slot. Pelvic patients are told 45 minutes before: they drink 500 mL of water and wait 30 minutes. When the machine runs ahead, ward and dormitory patients are called in, and patients living nearby get a "come early" message.
- **Arrivals, from the records.** With today's booking, patients arrive as the records show for an appointment at that hour: a median of about 10 minutes early in the morning and about 32 in the evening. With RAD-SMART, whose times are kept, they arrive as today's morning patients do (appointments 08:00–09:59, when the schedule still runs close to time), counted from the reporting time. The same random draw sets each patient's punctuality under every policy, so an early bird is early under both. Section 7.4 also shows RAD-SMART with today's arrival habits.
- **Other randomness.**
  - Lognormal session times.
  - 2% no-shows; 85% of patients act on an updated time.

- 0–3 urgent starts a day (Poisson 1.2, capped at 3), ready between 13:00 and 16:30.
- 2% imaging holds: half are repositioned at once (+8 min), half are treated 45 minutes later.
- The same random draws are used for every policy (common random numbers), so the comparisons are fair.

- **Waits** run from arrival to entering the treatment room, as in the department's records and in Munshi et al. [3]. They include any early arrival and the 20- or 45-minute preparation.
- **Independent rule checker.** A function separate from the solver re-verifies 8 hard rules on every plan: one patient at a time, the blood slot, the complex window, new starts by 17:00, MHRC slots, public transport by 21:00, the time windows set by the doctor, and accessories never double-booked with the CT simulator. On days built from standing slots it adds a ninth: every patient is in their standing slot.

7.2 Predicting machine-minutes

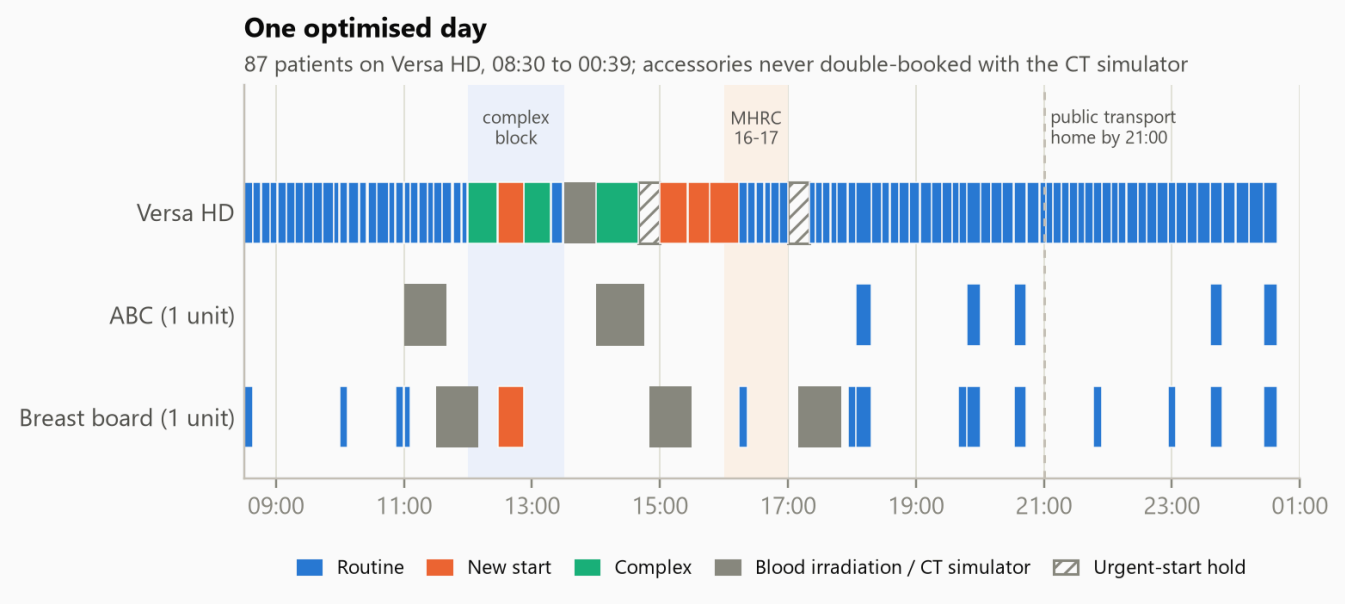


Error of predicted session length on 2,000 held-out synthetic sessions: lookup table of averages versus gradient boosting.

Model	Mean absolute error	Within 2 min	Within 3 min	P80 coverage (target 80%)
Lookup table of averages	2.62 min	50.9%	70.1%	80.4%
Quantile gradient boosting	**2.15 min**	**62.0%**	**78.3%**	78.5%

Trained on 12,000 synthetic historical sessions, the model learns effects the average table misses, such as first-fraction extra time, CBCT versus kV imaging, wheelchair and stretcher transfers, and faster delivery on the second machine. Plans reserve the model's expected minutes.

7.3 The optimised day



One optimised day on the Versa HD, 08:30 to 00:39. Complex cases stay inside 10:00–17:00, new starts finish by 17:00, MHRC patients are treated in their 16:00–17:00 slots, public-transport patients finish by 21:00, and the single breast board and ABC unit never clash with CT-simulator bookings.

On the PoC's 87-patient day, the machine-learning plan reached a proven optimum in 7.1 s and the average-table plan reached a proven optimum in 30.9 s on a laptop with the open-source HiGHS solver; neither left a patient unplanned. The model has about 4,800 variables and 1,662 constraints. Timings vary with the laptop's load. The independent checker found all 8 hard rules met on all 87 sessions. The plan also shows how the soft rules were traded off:

- 24 of the 24 sessions after 21:00 went to flexible patients (inpatients, dormitory residents and people living nearby).
- 13 of the 17 paying patients with a preferred time got it within 15 minutes.
- The three MHRC patients are treated at 16:14, 16:22, 16:30.
- 2 of the 3 complex cases sit in the 12:00–13:30 block.
- The last session is planned to end at 00:39.

A sample of the plan, with reason codes, is in Appendix C.

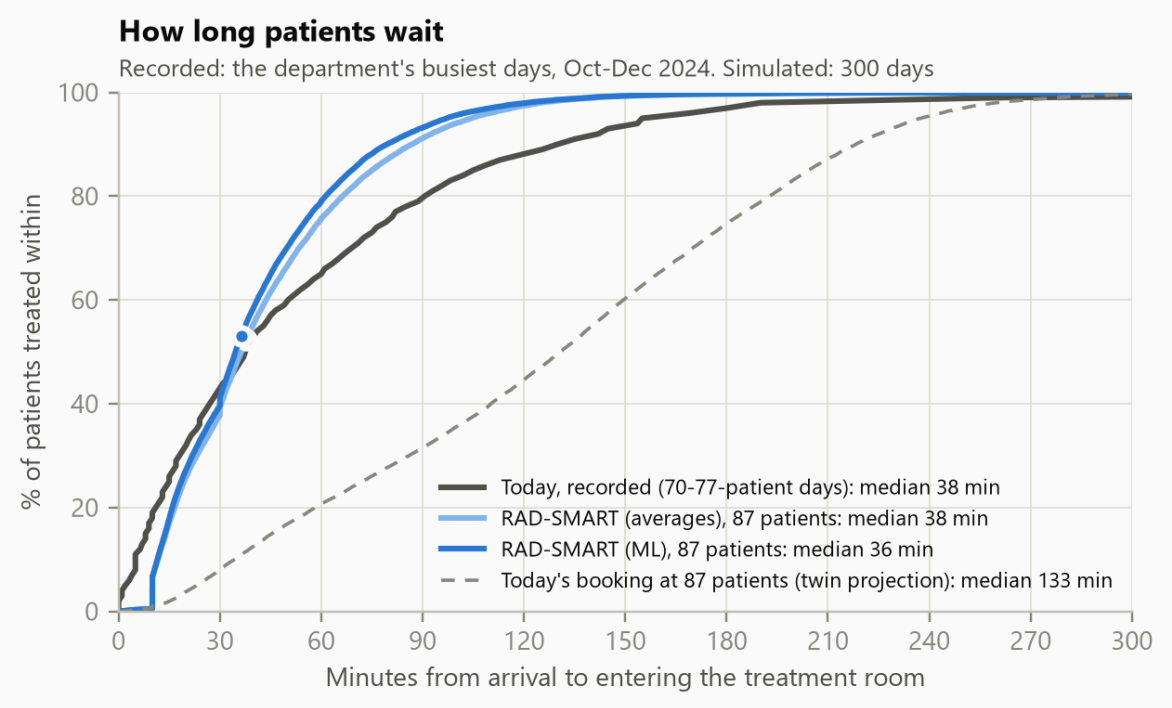
7.4 Waiting times over 300 simulated days

First, a check against the records. We replayed today's booking in the twin on days the size of each group of recorded days (40, 52, 64 and 74 patients) and compared the twin with the records:

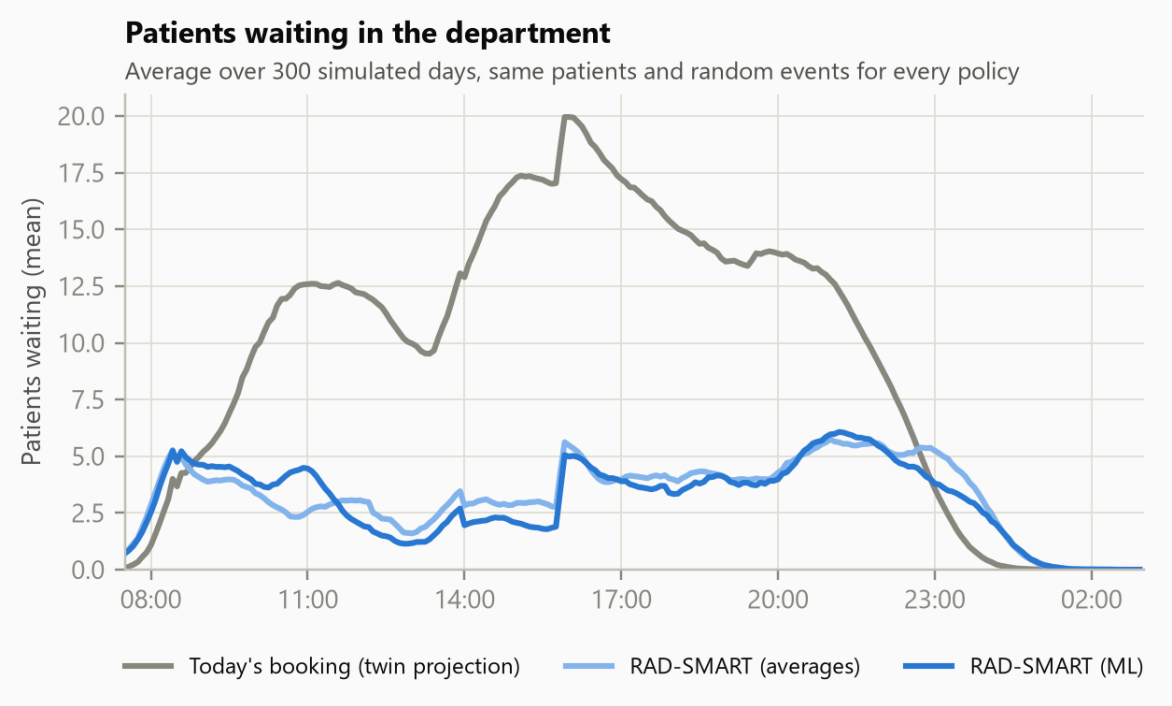
Day size (recorded days)	Median wait, records / twin (min)	90th-percentile wait, records / twin (min)	More than 1 hour late, records / twin
Up to 44 patients (7)	27.0 / 23.2	79.0 / 67.4	14.5% / 13.2%
45–59 patients (20)	32.0 / 30.3	98.0 / 81.6	21.8% / 17.1%
60–69 patients (18)	35.0 / 44.3	110.0 / 103.3	20.8% / 24.1%
70 or more patients (10)	37.5 / 63.2	130.0 / 144.2	27.4% / 38.6%
All 55 days	34.0 / 36.7	108.6 / 106.1	22.2% / 24.2%

Across all 55 days the twin reproduces today's waits closely: a 90th-percentile wait of 106 minutes against 109 recorded. On quieter days it under-predicts the long waits, because it leaves out delays such as machine QA

overruns, staff breaks and paperwork. On the busiest days it over-predicts them (63 against 38 minutes median at 74 patients): on busy days the department treats patients faster than our session-length estimates assume. We therefore quote today's figures from the records, and treat the twin's projection of today's booking at 87 patients as pessimistic.



Share of patients who entered the treatment room within a given number of minutes of arriving: the department's busiest recorded days, RAD-SMART on the 87-patient day, and the twin's projection of today's booking.



Mean number of patients waiting in the department through the day, in the twin.

Metric (mean of 300 simulated days)	Today, recorded (70–77-patient days)	Today's booking (twin projection)	RAD-SMART, averages	RAD-SMART, ML
Median wait, arrival to treatment room (min)	37.5	133.2	37.9	**36.3**
95% range of daily median wait (min)	–	81.5–206.8	27.6–61.3	24.4–65.4

Metric (mean of 300 simulated days)	Today, recorded (70–77-patient days)	Today's booking (twin projection)	RAD-SMART, averages	RAD-SMART, ML
90th-percentile wait (min)	130.0	210.2	82.2	**75.5**
Treated within ±15 min of appointment	30.1%	10.0%	48.9%	**52.9%**
Treated more than 15 min after appointment	53.1%	85.6%	33.3%	**27.1%**
Treated more than 1 hour after appointment	27.4%	66.3%	6.8%	**5.8%**
Median wait beyond the preparation time (min)	–	117.5	21.6	**19.3**
Mean time in department (min)	–	140.4	54.8	**52.3**
Last patient finishes (average)	–	23:36	01:01	01:04
Overtime after 01:00 (min per day)	–	0.1	10.8	11.7
New starts finishing after 17:00 (per day)	–	1.0	0.5	**0.3**
Complex cases outside 10:00–17:00 (per day)	–	0.8	0.3	**0.0**
MHRC patients missing the 17:00 bus (per day)	–	**0.2**	**0.2**	**0.2**
Public-transport patients finishing after 21:00 (per day)	–	1.3	**0.4**	0.6
Paying patients within 15 min of their preferred time	–	11.0%	44.4%	**47.5%**
Accessory-related delays (per day)	–	2.2	0.5	**0.3**
Delay to the blood-irradiation slot (min)	–	0.5	0.9	0.4
Urgent starts treated by 18:00	–	100%	100%	100%
Imaging holds (per day)	–	1.6	1.6	1.6

What changes. With 87 patients, 10 more than the busiest recorded day, RAD-SMART's median wait is about the same as on today's busiest days (36 against 38 minutes), because every patient is asked to report 20 minutes early (45 for pelvic patients) and some still come earlier. The difference is in the long waits and in punctuality: the 90th-percentile wait is 76 minutes against 130, 6% are treated more than an hour after their appointment against 27%, and 53% are treated within 15 minutes of it against 30%. Waits no longer build up through the day: the simulated median is about 36 minutes for patients arriving before 10:00 and 36 after 19:00, against 19 and 66 in the records (Section 2.1). Morning patients wait longer than today, mostly the 20-minute preparation they are asked to allow, and evening patients far less. In the twin, the waiting room holds at most about 6 people on average, against up to 20 with today's booking.

If patients keep today's habits. RAD-SMART's benefit assumes that, once times are kept, patients stop coming early for evening appointments. If they keep today's habits, the median wait rises to 44 minutes and the 90th percentile to 110, about the same as all recorded days and below the busiest (130), while only 5% are treated more than an hour late. Messages that give each patient a reliable time, and live updates when it changes, are how RAD-SMART earns that change. Bold marks the best value where lower or higher is clearly better.

7.5 How much capacity to hold for urgent starts

Holding machine time for same-day urgent starts and imaging re-treatments shortens waits, because a late-afternoon urgent patient does not push everyone behind them. But it ends the day later. We re-planned the same day with 40, 20 and 0 minutes held and simulated each plan 300 times.

Plan	Median wait (min)	90th percentile (min)	Within ±15 min	More than 1 h late	Last patient finishes	Overtime (min per day)	Urgent starts by 18:00
Today's booking (twin projection)	133.2	210.2	10.0%	66.3%	23:36	0.1	100%
40 min held (two 20-min holds)	36.3	75.5	52.9%	5.8%	01:03	11.7	100%
20 min held (one hold at 14:40)	41.2	81.2	51.5%	7.3%	00:50	6.9	100%
Nothing held	44.7	87.5	45.6%	9.3%	00:49	7.4	100%

Every RAD-SMART option keeps the 90th-percentile wait well below the busiest recorded days (130 minutes). Holding nothing finishes the day at 00:49, with a 90th-percentile wait of 88 minutes. Holding 40 minutes gives 76 minutes and finishes at 01:03. The department sets this as one rule in the Rule Studio and can change it by day of the week.

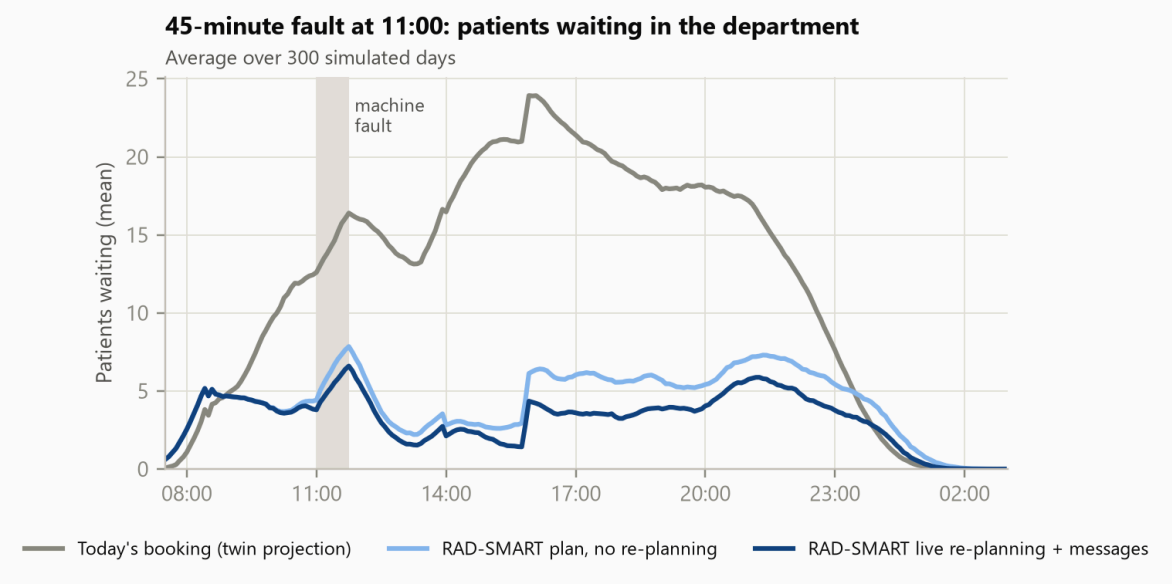
7.6 Machine downtime

We injected each of the department's two downtime cases into every simulated day and compared three responses:

1. Today's booking, unchanged (the twin's projection).
2. The RAD-SMART plan is kept but nobody is told.
3. RAD-SMART re-plans live under the department's rules and messages the patients who can still act on the change.

The fair test of re-planning is response 2 against response 3, which differ only in the re-plan.

45-minute fault at 11:00 (the up-to-60-minute rule).

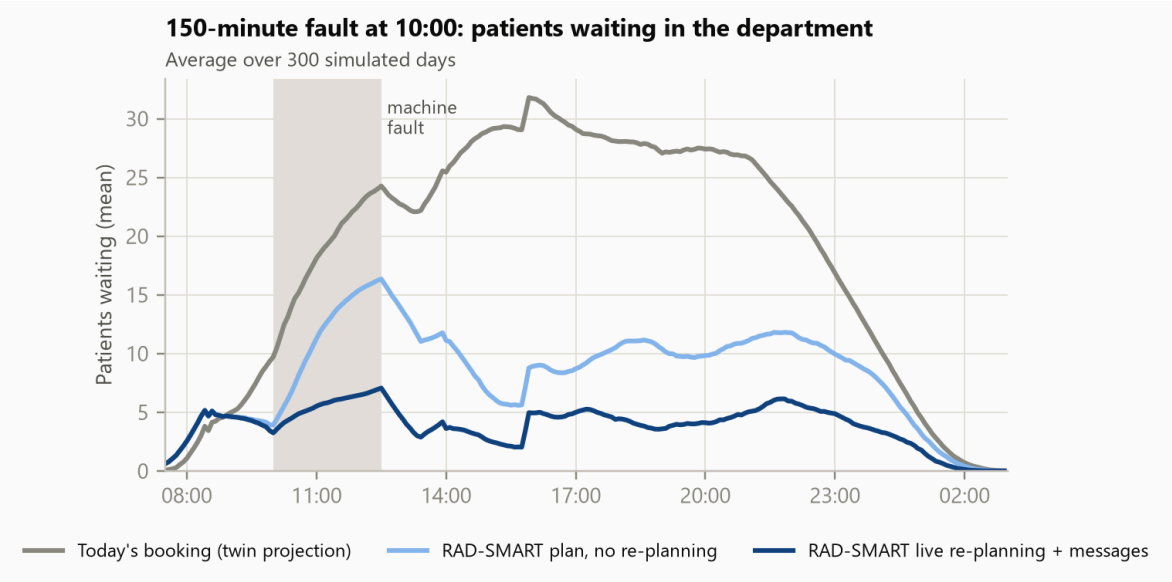


Mean patients waiting when the machine fails at 11:00 for 45 minutes.

Metric (mean of 300 simulated days)	Today's booking (twin projection)	RAD-SMART plan, no re-planning	RAD-SMART live re-planning + messages
Median wait (min)	177.3	52.6	**37.3**
90th-percentile wait (min)	256.8	102.2	**80.8**
Treated within ±15 min	8.4%	39.9%	**49.9%**
Treated more than 1 hour after appointment	74.1%	15.7%	**7.9%**
Mean time in department (min)	176.2	68.6	**54.6**
Last patient finishes (average)	00:17	01:18	01:07
Overtime after 01:00 (min per day)	2.7	22.4	13.4
New starts finishing after 17:00 (per day)	1.5	1.2	0.3
Complex cases outside 10:00–17:00 (per day)	1.0	0.1	**0.0**
Public-transport patients finishing after 21:00 (per day)	2.6	0.9	0.5
MHRC patients missing the 17:00 bus (per day)	0.2	0.3	0.2
Patients moved to another day, after approval (per day)	0.0	0.0	1.0

The re-plan took **10.1 seconds**, within the 30-second target. Following the up-to-60-minute rule, nobody was sent home: the machine is back at 11:45 and every patient is treated today, with the plan running 8 minutes past its planned end. Patients already waiting are treated first and told in person. It moved 45 patients and messaged 40, those who had not yet reported and whose time changed by at least 5 minutes. The day's one new start that could no longer finish by 17:00 moved to the next working day, for a technologist or doctor to approve.

150-minute fault at 10:00 (the over-60-minute rule).

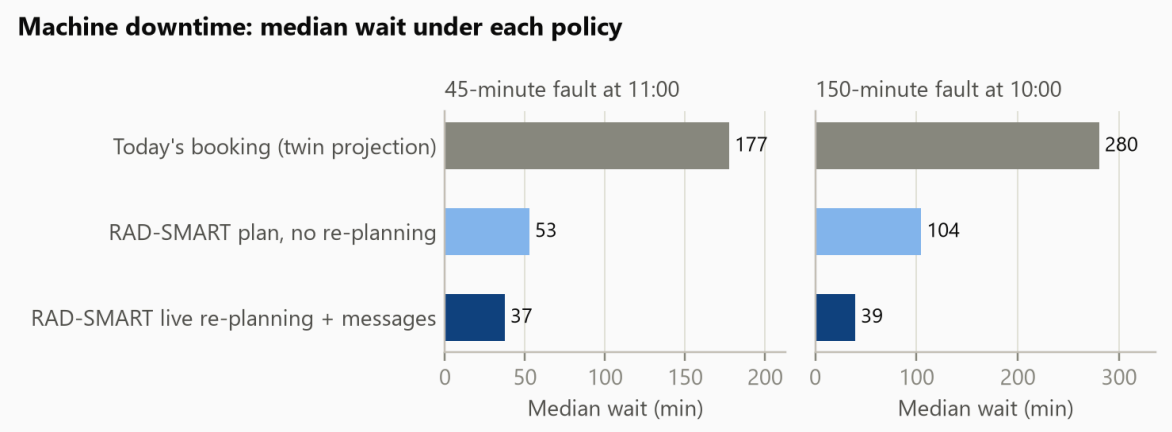


Mean patients waiting when the machine fails at 10:00 for 150 minutes.

Metric (mean of 300 simulated days)	Today's booking (twin projection)	RAD-SMART plan, no re-planning	RAD-SMART live re-planning + messages
Median wait (min)	280.2	104.4	**39.4**

Metric (mean of 300 simulated days)	Today's booking (twin projection)	RAD-SMART plan, no re-planning	RAD-SMART live re-planning + messages
90th-percentile wait (min)	357.7	210.1	**110.0**
Treated within ±15 min	6.8%	15.1%	**47.0%**
Treated more than 1 hour after appointment	82.8%	58.7%	**13.6%**
Mean time in department (min)	265.4	125.4	**66.2**
Last patient finishes (average)	01:46	01:56	01:46
Overtime after 01:00 (min per day)	47.0	55.8	45.8
New starts finishing after 17:00 (per day)	3.1	3.0	**0.3**
Complex cases outside 10:00–17:00 (per day)	1.5	1.2	**0.0**
Public-transport patients finishing after 21:00 (per day)	5.1	2.5	**0.5**
MHRC patients missing the 17:00 bus (per day)	0.2	0.2	0.2
Patients sent back or moved to another day, after approval (per day)	0.0	0.0	3.0

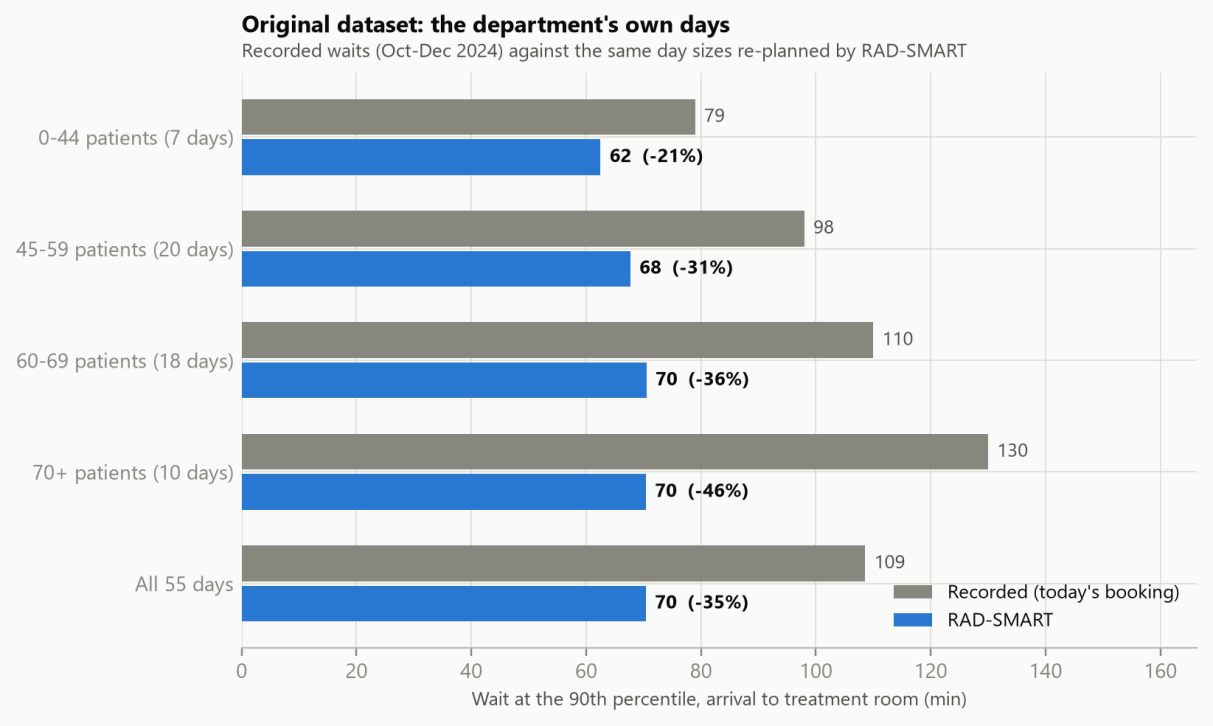
The re-plan took **13.9 seconds**. The over-60-minute rule applies: the machine is back at 12:30, and 60 minutes of overtime and the gaps in the plan could not absorb 39 machine-minutes. RAD-SMART therefore proposed sending back one patient who had not yet set out, told not to travel, with the fraction added at the end of the course. It moved the two new starts that could no longer finish by 17:00 to the next working day. Everyone else is treated today: 63 patients moved and 61 messaged. Without re-planning, the RAD-SMART plan runs to about 01:56; new starts and complex cases finish late, and public-transport patients miss their buses. With re-planning, the median wait is 39 minutes. The price of keeping nearly everyone in the day, as the rule asks, is a long evening: the day ends at about 01:46, with 46 minutes of overtime after 01:00.



Median wait under each response, for both faults.

7.7 The department's own days, re-planned

The tests above use synthetic days. To ask what RAD-SMART would have done on the department's own days, we took the 55 recorded days in four groups by size and re-planned a day of each size in the twin, with patients arriving as early or as late as the records show for each hour. "Without" is what was recorded; "with" is RAD-SMART. Each group's result is weighted by its number of days. This is a replay in the twin, not a live trial, and Section 7.4 shows how close the twin comes to the records.



The department's own days. Grey: the recorded 90th-percentile wait. Blue: the same day sizes re-planned by RAD-SMART.

Day size (recorded days)	90th-percentile wait, recorded → RAD-SMART	Within ±15 min	More than 1 hour late
Up to 44 patients (7)	79 → **62 min** (−21%)	30.5% → **41.4%**	14.5% → **4.7%**
45–59 patients (20)	98 → **68 min** (−31%)	28.9% → **46.3%**	21.8% → **6.1%**
60–69 patients (18)	110 → **70 min** (−36%)	27.9% → **51.7%**	20.8% → **5.8%**
70 or more patients (10)	130 → **70 min** (−46%)	30.1% → **53.6%**	27.4% → **5.5%**
All 55 days	108.6 → **70.4 min** (−35%)	28.9% → **49.5%**	22.2% → **5.7%**

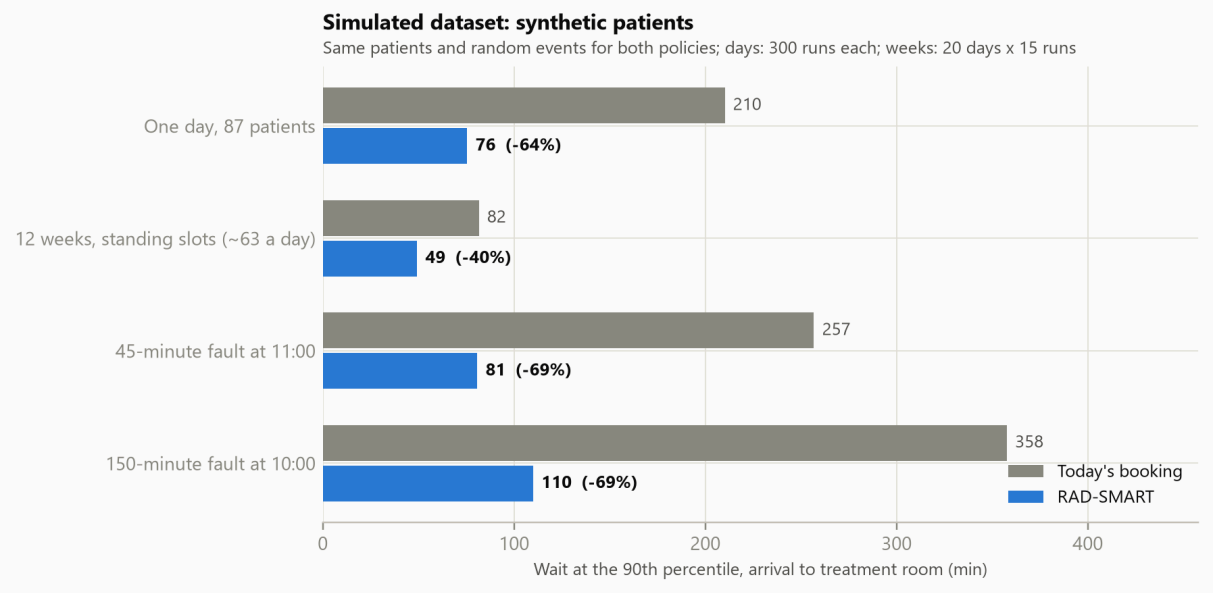
The busier the day, the bigger the gain: today the long waits grow with the size of the day, but with RAD-SMART they stay near 70 minutes. The median wait moves from 34 to 30 minutes, because patients are still asked to report 20 minutes early (45 for pelvic patients).

7.8 Twelve weeks with standing slots

The department keeps each patient's slot for the whole course, so the fairest test follows courses over weeks. Starting from an empty machine, new patients arrive every working day (a Poisson stream sized for about 70 on treatment once courses overlap) and each one is booked twice, on the same start date:

- **today's booking:** a time drawn from the department's recorded spread of appointment times, inside the patient's own window, kept for the course; nothing checks that the machine is free;
- **RAD-SMART:** a new-start slot before 17:00 and a standing slot that is free on every treatment date and leaves the 150-minute new-start reserve (Section 6.4). When no new-start gap is left that day, the start moves to the next working day with one, and both policies use that date.

After seven weeks, when courses overlap fully, 20 working days are played out in the twin under both policies, 15 times each, with the same patients and the same random events.



Both datasets of synthetic patients: one 87-patient day, 12 weeks with standing slots, and the two breakdowns. Grey: today's booking. Blue: RAD-SMART.

Metric (20 days × 15 runs)	Today's booking	RAD-SMART, standing slots
Patients per day	62.7 (58–68)	62.8
Median wait (min)	40.3	**19.9**
90th-percentile wait (min)	81.6	**49.3**
Treated within ±15 min	25.8%	**35.6%**
Treated more than 1 hour after appointment	18.6%	**3.6%**
Mean time in department (min)	55.7	**37.2**
Last patient finishes (average)	21:55	00:07
Overtime after 01:00 (min per day)	0.0	8.3
Patients finishing after 21:00 who are not flexible (per day)	0.2	6.2
Public-transport patients finishing after 21:00 (per day)	0.1	0.2
New starts finishing after 17:00 (per day)	0.3	**0.0**
Patients treated within 15 min of a time they asked for	**31.4%**	3.9%
New patients starting later than the requested date	–	21.9% (0.42 working days on average)
Hard rules broken (independent check, 9 rules, 20 days)	not checked	**0**

What standing slots cost. The gains in waiting are clear, and so are the costs. The evening runs about two hours later, still inside the department's hours, because patients are spread out instead of being given the same time and 150 minutes before 17:00 is kept for new starts. One new patient in five starts later, by less than half a working day on average. And a patient who asks for a particular time rarely gets it: today they are given that time but then wait in the queue, while a standing slot must be free on every day of the course. A slot-change request moves them when a nearer slot frees up, as courses end every day. These are the department's choices to weigh, and the pilot measures them.

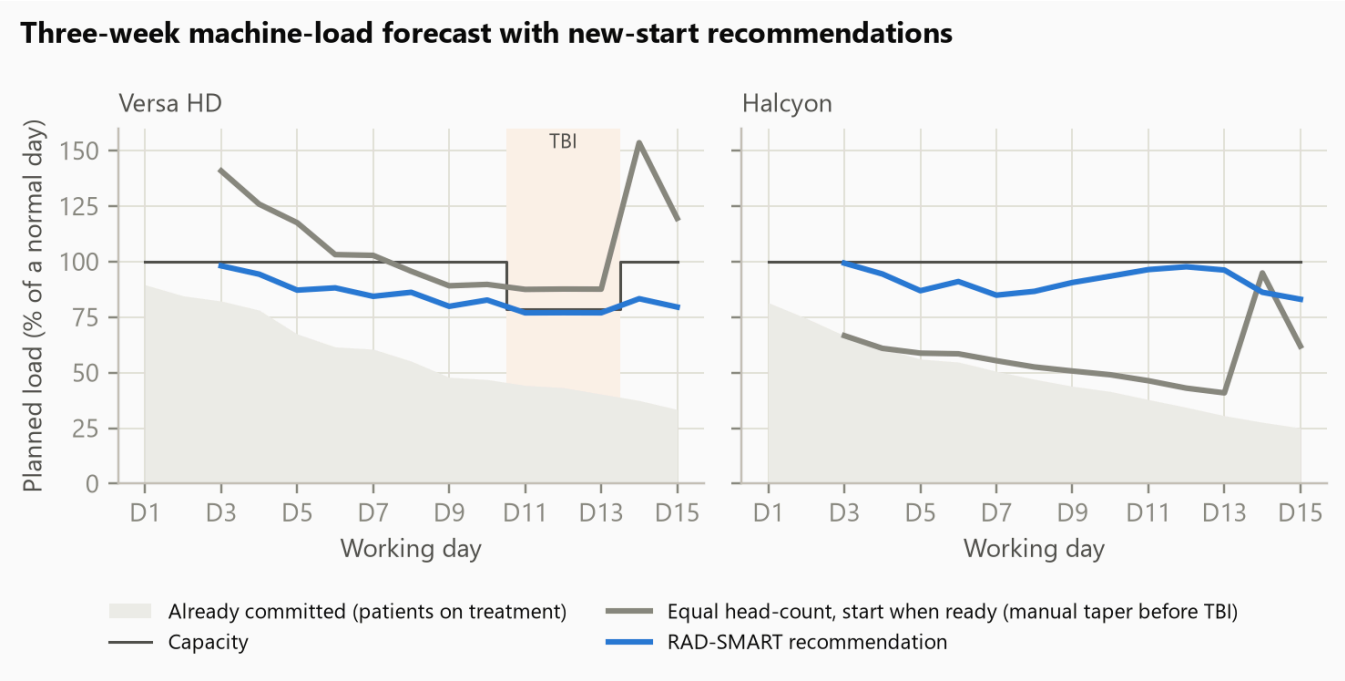
7.9 Two machines, three weeks and a TBI course

We tested the forecast on a synthetic two-machine department:

- 79 patients on treatment on the Versa HD and 111 on the second machine (a Halcyon, as a placeholder until the capability matrix is agreed);

- 99 patients in the planning pipeline over three weeks;
- a TBI course on the Versa HD on days 11–13, taking 180 minutes a day.

RAD-SMART was compared with the common practice of splitting new patients equally by head-count and starting each as soon as they are ready, with a manual taper before TBI (at most 2 new starts a day on days 6-13).



Three-week planned load per machine. The grey area is load already committed by patients on treatment; capacity drops on the TBI days.

Result	Equal head-count with a manual taper	RAD-SMART recommendation
Peak planned load, Versa HD	154%	**98%**
Peak planned load, Halcyon	95%	**99%**
Versa HD on the TBI days (share of the reduced capacity)	111%	**98%**
Decision days over capacity, Versa HD	10 of 13	**0**
Average working days from plan ready to first fraction	1.7	**0.0**
Pipeline patients still waiting at the end of the horizon	0	**0**

Splitting by head-count overloads the slower Versa HD while the second machine sits under-used, which is the "forty simple versus thirty complex" point made in the problem statement. The manual taper then delays starts and releases a backlog after TBI. Balancing by predicted minutes keeps both machines at their target, fits the TBI days exactly and starts every patient when their plan is ready. The advisor's three weeks:

Day	Versa HD load	Finishing	New starts	Halcyon load	Finishing	New starts
Mon (day 1)	90%	5	–	81%	9	–
Tue (day 2)	85%	2	–	74%	10	–
Wed (day 3)	82%	4	5	67%	7	12
Thu (day 4)	78%	5	3	61%	6	8
Fri (day 5)	67%	5	3	56%	2	4
Mon (day 6)	62%	1	5	55%	5	4

Day	Versa HD load	Finishing	New starts	Halcyon load	Finishing	New starts
Tue (day 7)	61%	4	2	51%	4	2
Wed (day 8)	55%	6	3	47%	4	3
Thu (day 9)	48%	1	1	44%	3	5
Fri (day 10)	47%	2	2	42%	4	5
Mon (day 11) · TBI	44%	1	1	38%	4	5
Tue (day 12) · TBI	43%	2	2	34%	4	6
Wed (day 13) · TBI	40%	2	5	31%	3	4
Thu (day 14)	37%	4	5	28%	3	0
Fri (day 15)	33%	2	4	25%	0	0

"Load" is the share of a normal day's capacity already committed by patients on treatment; "new starts" is the recommendation that fills the gap up to the target, after the TBI slots. Days 1 and 2 are already booked, because decisions are made two working days ahead.

7.10 Limitations of the PoC

- **Synthetic patients, calibrated behaviour.** The rules are the department's and the twin's booking pattern and arrival habits come from its records, but session lengths and the patient mix are our estimates. The size of the real improvement depends on how much of today's waiting comes from scheduling rather than from other causes, such as transport, machine QA or paperwork.
- **The twin is not perfect.** It reproduces a typical recorded day closely on median wait and punctuality but under-predicts the longest waits (93 against 110 minutes at the 90th percentile), so RAD-SMART's simulated tail may be optimistic by a similar margin. On busy days it over-predicts today's waits, which is why we compare with the records rather than with the twin's projection.
- **Standing slots from an empty machine.** The 12-week test starts from an empty machine with synthetic arrivals. The pilot starts from today's patients, mid-course with their current times: they keep those times, and only new courses get standing slots, so the full benefit builds up over about seven weeks as courses turn over.
- **The original dataset is replayed, not re-run.** Section 7.7 re-plans days of the recorded sizes in the twin with the recorded arrival habits; it does not re-run the exact patients of those days, whose session lengths were never recorded.
- **Patient behaviour.** The benefit assumes that patients stop coming early once times are kept; with today's habits the 90th-percentile wait is 110 minutes rather than 76. It also assumes that 85% of patients act on updated times. The pilot will measure both.
- **One machine, one laptop.** The day simulation covers one machine; the two-machine test covers the forecast, not simultaneous live sequencing, which is part of the build sprint. Solver times were measured on a laptop with an open-source MILP solver, and a plan can stop at its time limit with a small proven gap. Production will use OR-Tools CP-SAT, which is well suited to interval scheduling.

For these reasons the pilot target (at least 25% lower 90th-percentile wait than the records' 109 minutes) is set below the PoC results: 35% lower on the department's own days re-planned, 40% lower over 12 synthetic weeks with standing slots, and 42% lower than the busiest recorded days on the 87-patient day.

8. KPIs and expected impact

8.1 Measurement framework

Type	KPI	Definition and source	Pilot target
Primary	90th-percentile gross waiting time (WTG)	Linac room entry minus arrival (QR check-in and tablet tap), all sessions, per day	≥ 25% lower than baseline by day 90 (records: 109 min)
Secondary	Median WTG	As above	Not higher than baseline (records: 34 min)
Secondary	Punctuality	% of sessions starting within ±15 min of the appointment, and more than 1 hour after it	±15 min: from 29% to ≥ 45%; over 1 hour late: from 22% to ≤ 10%
Secondary	Evening build-up	Median WTG after 19:00 against before 10:00	Evening no more than 10 min longer (records: 66 against 19 min)
Secondary	New starts on time	% of new starts finished before 17:00	100%
Secondary	Complex cases in window	% of complex sessions inside the senior-staff window	100%
Secondary	MHRC patients on the bus	% of MHRC patients finished in time for the 17:00 bus	100%
Secondary	Public-transport patients home in time	% finished by 21:00	100%
Secondary	Urgent starts on the same day	% of urgent starts treated by 18:00	100%
Secondary	Daily overtime	Minutes after regular end	Not above baseline
Secondary	Scheduling conflicts and accessory delays	Count per week	≥ 80% fewer
Secondary	Prediction accuracy	Mean absolute error; % within 3 min	MAE ≤ 2.5 min
Secondary	Occupancy forecast error	Predicted versus actual daily machine-minutes	≤ 10%
Secondary	RTT scheduling effort	Self-reported minutes per day spent on appointments	≥ 50% lower
Balancing	Throughput	Patients treated per day	Not lower than baseline
Balancing	End of the day	Time the last session ends	As the department chooses (Section 7.5)
Balancing	Downtime handling	Patients deferred per fault, and approvals	Only as the downtime rules allow
Balancing	Equity	Median wait by group: elderly, transport-dependent, language, paying versus non-paying	No group worse off
Balancing	Human control	Override rate and reasons; plans approved without edits	Tracked; reviewed weekly
Balancing	Patient experience	Short survey in the patient's language (predictability, stress)	Improvement over baseline

8.2 Impact estimate (illustrative, conservative)

- **Per patient.** The gain is concentrated where it hurts. Today one visit in ten ends in a wait of more than 109 minutes across all recorded days, and 130 on the busiest; an evening patient waits about 66 minutes. In the PoC, on a day busier than any recorded, these fall to 76 and about 36 minutes. For an evening patient on a 25-fraction course, that is about 12 hours saved, and roughly as much again for the caregiver who comes along.

- **Per machine per year.** On busy days the mean wait is 55 minutes in the records and 41 in the PoC. 14 minutes × 80 patients × 250 working days ≈ **4,600 patient-hours**, about twice that including caregivers, before counting the early arrivals that reliable times make unnecessary.
- **Capacity.** Balanced loading, fewer idle gaps and faster backfilling of no-shows free up machine time. As an illustration only, not a PoC result: if smarter scheduling recovered 5% of machine time across India's ~823 machines [6], that would equal about 41 machines, or roughly ₹800–1,000 crore of equipment at recent public-sector prices (two linacs at AIIMS Jhajjar cost about ₹50 crore [21]).
- **Scale.** The rules are configuration, not code, so another department can adopt RAD-SMART by describing its own rules in the Rule Studio. The National Cancer Grid's 370+ member institutions, which serve about 60% of India's cancer patients, are the natural path to scale.

9. Implementation plan

9.1 Timeline

Phase	Dates	Outcome
Round 1 submission	by 25 Sep 2026, 23:59	Idea, deck, PoC evidence, this report
Evaluation and shortlisting	26 Sep – 3 Oct 2026	
Build sprint (5 weeks)	5 Oct – 8 Nov 2026	Working prototype on synthetic patients, calibrated on the department's records and validated with the RTT team
Top 30 announced	by 14 Nov 2026	
Grand finale, IIT Bombay	28 Nov 2026	Live demonstration
Approvals	Nov 2026 – Jan 2027	HoD, ethics/QI, IT, messaging registrations
90-day pilot	from approval (target Q1 2027)	KPI evidence
Scale-up	after pilot	Halcyon, OIS integration, other NCG centres

9.2 Build sprint plan (5 Oct – 8 Nov)

Week	Build	Doctor partner's role
W1 (5–11 Oct)	Data model and Excel import; rule configuration (YAML) with versioning; port the optimiser to OR-Tools CP-SAT; synthetic data generator calibrated with the department's averages	Agree the departmental estimate table (minutes per technique, first fraction, accessory); settle the remaining open items (Appendix B)
W2 (12–18 Oct)	RTT console: day timeline, live queue board, approve / edit / lock, override with reason, audit log; live re-planning; accessory tracking with QR tags	Walk-through with RTTs; check that the screens fit the real workflow
W3 (19–25 Oct)	Sarvam integration: copilot ("why" answers, Rule Studio), WhatsApp / SMS / voice templates in Kannada, Malayalam, Hindi and English, with recorded Tulu voice notes, spoken "running late" replies, OCR import of a paper register	Review message wording; arrange native-speaker checks of translations and the Tulu recordings
W4 (26 Oct – 1 Nov)	Two-machine allocation; capacity forecast and new-start advisor; KPI and equity dashboard; digital-twin what-if screen	Validate forecast logic against real experience; agree KPI definitions
W5 (2–8 Nov)	Hardening, independent rule checker, test suite, usability test with RTTs on synthetic data, demo video, final submission	Final clinical-fit review; co-present

Progress at 24 October. After the department's answers of 24 October we moved the console into CARE (Section 6.10). Built and tested end to end on synthetic data: the RADSMART button on the visit page, the intake form and checklist, the proposal and approval of a course's new-start and standing slots, slot-change requests, the day board with arrival and room times, the breakdown response under the new rules with drafted messages, and Insights. Still to build: the Sarvam copilot and message sending, Excel and paper import, the Rule Studio, QR check-in and accessory tags, the two-machine screens and the port to CP-SAT.

9.3 Finale demonstration (28 Nov)

1. In CARE, open a synthetic patient's visit page, click **RADSMART**, enter the course and tick the priority checklist. RAD-SMART proposes a new start before 17:00 and a standing slot, and the doctor approves.

2. On the technologist's day board, show the rule check, the timeline and the patients' recorded times, and approve the day. Load a list from Excel and a photo of a paper register.
3. A patient's phone receives the reporting time in Kannada script with a recorded Tulu voice note, and a voice call plays in Malayalam.
4. Report a 45-minute and then a 150-minute breakdown. Each proposal follows the department's downtime rule, a technologist approves it, and messages go out.
5. A patient says "I'm stuck in traffic" by voice, and the queue adjusts.
6. Ask the copilot "why was this patient moved?" and change a rule in plain language, with diff, approval and re-plan.
7. Show the two-day-ahead new-start advice for two machines and the KPI dashboard.

9.4 90-day pilot design

Days	Phase	What happens
1–14	Baseline	QR check-in and room entry and exit taps start, adding session durations to the appointment, arrival and treatment times the department already records, with no change to scheduling. This refreshes the baseline in the records and collects duration data. RAD-SMART plans from the department's estimate table until enough durations exist
15–35	Shadow mode	RAD-SMART plans every evening. RTTs keep scheduling as usual and compare. Prediction accuracy and plan acceptability are measured
36–90	Live on the Versa HD	The doctor approves every new course and its slots; a technologist or doctor approves the day's plan and every breakdown response. Patients receive times and live updates. Re-planning is live. The model is retrained weekly

- **Analysis.** An interrupted time series (segmented regression) of the daily 90th-percentile wait, with weekly statistical process control charts, plus before/after comparison of the secondary and balancing KPIs.
- **Stopping rules.** If a hard rule is broken in practice, or balancing KPIs worsen for two consecutive weeks, the department reverts to manual mode, reviews the cause, and resumes only when it is fixed.
- **After the pilot.** Extend to the second machine (early 2027) once its capability matrix is agreed, add multi-machine live sequencing, and add a read-only OIS feed.

9.5 Team and roles

Role	Who	Time
Team leader, doctor partner and clinical owner	Dr Akshay Dinesan	about 2 h per week in the sprint; pilot sponsor
Technical lead	Abhinand T M	full sprint
Lead engineer	P S Raghotham Rao	full sprint
Doctor partner	Dr Shirley Lewis Salins	sprint and pilot
Doctor partner	Dr Umesh Velu	sprint and pilot
RTT champion (pilot)	[to be nominated by the department]	about 1 h per week
Approvers	doctor (courses and slots); technologist or doctor (the day's plan and breakdowns)	daily during the live phase
Medical physicist	[to be nominated]	machine capability matrix; rule review
Hospital IT	[to be nominated]	server, network, security review

10. Requirements

10.1 Functional requirements

ID	Requirement
FR01	Import the daily treatment list from Excel or CSV (or an OIS export) and validate it
FR02	Import from a photo of a paper register by OCR, with human confirmation of every row
FR03	Configure machines, staff windows, blocks, accessories, priorities and weights in a Rule Studio with versions, diffs and rollback
FR04	Predict P50 and P80 machine-minutes per session, falling back to average tables
FR05	Generate the evening-before plan per machine with hard rules guaranteed
FR06	Provide reason codes for every placement and plain-language "why" answers
FR07	Let the RTT review, edit, lock and approve; log every override with its reason
FR08	Notify patients in their chosen language and channel (WhatsApp, SMS, voice), with an optional caregiver copy
FR09	Capture QR check-in and room entry and exit times
FR10	Show a live queue board with anonymised tokens
FR11	Handle events (delay, fault, no-show, urgent add-on, patient running late) with a re-plan proposal within 30 s
FR12	Re-plan with minimum disruption and targeted notification rules
FR13	Track accessories and detect conflicts, including CT-simulator bookings
FR14	Allocate across multiple machines by capability and predicted minutes
FR15	Provide a 3-week load forecast and a 2-day-ahead new-start recommendation per machine
FR16	Run digital-twin what-ifs (extra new starts, service day, staff change)
FR17	Provide a KPI and equity dashboard
FR18	Export the audit trail
FR19	Provide a printable fallback schedule and a manual-mode switch
FR20	Retrain the model weekly with a drift report; human approval before a new model goes live
FR21	Apply the department's downtime rules (15 min to 48 h: up to 60 min everyone is treated with overtime; over 60 min patients are sent back as the expected downtime needs; new starts wait; patients not yet travelling are told to postpone) and route the response to a technologist or doctor for approval
FR22	Give every patient a reporting time and preparation instructions, such as bladder filling for pelvic patients
FR23	Plan TBI courses and adjust new-start capacity around them
FR24	Model accessory reservations for the linac and release them to the CT simulator when unused
FR25	Give each course a new-start slot before 17:00 and a standing slot kept until the course ends; handle a patient's slot-change request with a proposal and an approval
FR26	Run inside the hospital information system (CARE in the build sprint) with a RADSMART button on the patient's visit page, and an intake form with the prescription, treatment days and a tick box for every hard and soft rule
FR27	Let the technologist record arrival, room entry and exit, machine status and issues; let the doctor enter and approve courses

10.2 Non-functional requirements

- **Performance.** Evening plan for 90 patients per machine in under 60 s; live re-plan in under 30 s; console responses under 1 s.
- **Reliability.** Runs on the hospital LAN if the internet is down, with messages queued and an SMS or voice fallback. Nightly encrypted backup and a printable fallback list.
- **Security and privacy.** On the hospital's server or in an Indian cloud region; role-based access checked on the server; TLS; pseudonymised IDs in analytics; no personal identifiers sent to the language model; data minimisation and retention limits.
- **Usability.** Designed for tablets with large touch targets. Staff interface in English and Kannada, and patient channels in Kannada, Tulu (recorded voice), Malayalam, Hindi and English. Training takes under 30 minutes.
- **Auditability.** Append-only log; every plan reproducible from its inputs, rule version and model version.
- **Portability.** All department rules are configuration; open-source core components.

10.3 Minimum dataset (operational only)

Field	Example	Source
Patient token (pseudonymous)	P034	Generated locally
Machine and eligible machines	Versa HD	Physics matrix
Technique and site group	VMAT, pelvis	Treatment card
Fraction number / total	12 / 30	Treatment card or OIS
First fraction, imaging type	yes; CBCT	Treatment card
Accessories	breast board, ABC	Treatment card
Mobility	wheelchair	RTT
Logistics flags	inpatient, MHRC (hospital bus), dormitory, lives nearby, public transport with latest return time	Reception or RTT
Preparation	pelvic bladder-filling protocol (reports 45 min early)	Treatment card
Priority flags	elderly, paying (if the department keeps this policy)	Registration
Requested or usual time	09:00	Patient or RTT
Language, channel, caregiver contact	Tulu, voice	Patient (stored on the hospital server only)
Timestamps	arrival, room entry, room exit	QR check-in, tablet or OIS

No diagnoses, doses, images or clinical notes are collected.

10.4 Technology stack

Layer	Choice
Optimisation	Google OR-Tools CP-SAT (production); HiGHS via SciPy (PoC)
Machine learning	scikit-learn quantile gradient boosting (LightGBM optional)
Simulation	Python and NumPy discrete-event twin
Backend	Python; a Django plugin inside CARE (Django REST framework, PostgreSQL, Celery)
Frontend	A React plugin loaded into CARE's web app (Vite Module Federation), for desktops and tablets; a TV queue board

Layer	Choice
Language and voice AI	Sarvam-105B, Bulbul v3, Saaras v3, Sarvam Translate / Mayura, Sarvam Vision; staff-recorded Tulu voice clips
Agent orchestration	LangGraph with an allow-listed tool set; template fallback
Messaging	WhatsApp Business Platform (utility templates); SMS through a DLT-registered gateway; cloud telephony for voice calls
Deployment	One Docker image (CARE's production build with both plugins, nginx, gunicorn). Live now on Microsoft Azure: Container Apps, Azure Database for PostgreSQL with daily backups, Container Registry, a daily job. For the pilot: the same image on the hospital's server (Docker Compose) or in an Indian Azure region
Integration (phase 2)	Read-only HL7 v2 or FHIR R4 feed from the OIS (MOSAIQ / ARIA)

10.5 Hardware and pilot budget (indicative, to be confirmed with quotes)

Item	Estimate (₹)
On-premise server with UPS (or an existing hospital VM at no cost)	1,00,000 – 1,50,000
3 Android tablets (reception check-in, machine console, RTT lead)	45,000 – 60,000
TV display for the queue board (if not available)	20,000 – 30,000
QR label printer, labels and patient cards	10,000 – 15,000
Messaging for 90 days (WhatsApp, SMS, voice); confirm current rate cards	15,000 – 40,000
Sarvam API usage (may be covered by partner credits)	10,000 – 25,000
Total	**about 2.0 – 3.2 lakh**

10.6 Approvals and registrations

1. Head of department approval for the pilot and the RTT workflow change.
2. Institutional Ethics Committee review as a quality-improvement study (exemption or expedited review, as the committee decides).
3. Hospital IT and security sign-off for the server, network access and data flows.
4. A patient information notice and consent for messaging (channel, language, caregiver copy), aligned with the DPDP Act 2023 and Rules 2025.
5. Meta business verification of the hospital's WhatsApp number, with approval of each message template.
6. DLT registration of the sender ID and SMS templates with a telecom operator, as TRAI requires.
7. Physics sign-off of the machine capability matrix and accessory list.

11. Risks, ethics and compliance

11.1 Risk register

#	Risk	Likelihood / impact	Mitigation
1	Staff do not adopt the plan or override it constantly	Medium / High	Co-design with RTTs, shadow mode first, one-tap override, weekly review of override reasons
2	Patients do not read or act on messages	Medium / Medium	Voice calls, caregiver copy, own language, reception reminder; plan is robust at 85% compliance
3	Duration predictions are poor at first	Medium / Medium	Start with average tables (already most of the gain); P80 buffers; weekly retraining
4	Session durations are not recorded today, and capture may be incomplete	High / High	Start from the department's estimate table; QR check-in and tablet taps from day 1; OIS timestamps where available; completeness tracked (target ≥ 90%)
5	A hard rule is broken in practice	Low / High	Solver guarantees plus an independent checker; human approval; stopping rule
6	The language model produces a wrong or unsafe answer	Medium / Medium	It never writes schedules; answers grounded in reason codes; fixed, approved message templates; logging
7	Privacy breach or DPDP non-compliance	Low / High	On-premise; pseudonymisation; no identifiers to external APIs; consent; access control; audit
8	Internet or WhatsApp outage	Medium / Low	LAN operation; SMS or voice fallback; printed list
9	Priority rules disadvantage some groups (for example, paying-patient preference)	Medium / High	Transport limits are hard rules; equity dashboard; weights set openly by the department and reviewed monthly
10	Scope creep into clinical decisions	Low / High	Urgency, technique and eligibility always entered by clinicians; the system only advises on counts and timing
11	Solver too slow on large or multi-machine days	Low / Medium	Two-stage decomposition; time limit with best plan and quality gap shown; CP-SAT
12	Other changes during the pilot confound the result	Medium / Medium	Interrupted time-series design, a baseline period, balancing measures, honest reporting
13	Tulu speakers are not reached well, because no speech model covers Tulu	Medium / Medium	Kannada-script text with staff-recorded Tulu voice notes; Kannada speech on request; reception reminder; recordings reviewed monthly
14	Holding capacity for urgent starts makes the day end later	Medium / Low	The department chooses the hold (Section 7.5); unused holds are used for catch-up
15	Standing slots end the day later, delay some new starts and give fewer patients the time they ask for (Section 7.8)	Medium / Medium	The department sets the new-start reserve and the evening rules; slot-change requests move patients as slots free up; equity dashboard

11.2 Ethics and regulation

- **Hackathon scope.** RAD-SMART does not diagnose, recommend treatment, score clinical risk or interpret medical data. Its only "prediction" is how many minutes a machine session will take.
- **Medical device status.** Its intended purpose is operational scheduling. We will confirm with the institution's regulatory team that it falls outside software-as-a-medical-device classification.
- **Data protection.** The DPDP Rules 2025 were notified on 13 November 2025, with the main obligations applying from May 2027 [18]. The pilot is designed to comply from the start: clear notice, consent for

messaging, purpose limitation, data minimisation, retention limits, breach handling and data-principal rights.

- **Fairness.** The problem statement lists paying patients among the priority categories. RAD-SMART makes every priority an explicit, visible weight chosen by the department, reports waits by group, and treats a rule that makes any group worse off as a problem to review.
- **Responsible AI.** Every AI output is logged with its model version and inputs. People can see why a decision was proposed and can override it, in line with Indian guidance on ethical AI in healthcare.
- **Consent for messaging.** WhatsApp messages are sent only to patients who opt in. SMS uses registered templates in line with TRAI's rules on commercial communications.

12. Innovation summary

1. **Machine-minutes, not head-counts.** Quantile predictions (P50/P80) drive every decision, as the problem statement asks.
2. **Guaranteed rules.** Hard constraints are enforced by an optimiser and re-checked independently. They are never left to an AI model's judgement.
3. **Two-stage coarse-to-fine optimisation.** A 70–90-patient day is solved to a proven optimum, or within a stated gap, in under a minute with open-source tools.
4. **Minimum-disruption live recovery under the department's downtime rules.** Re-planning treats the people already waiting first, moves as few others as possible, never calls anyone earlier than they can arrive, and messages only people who can act.
5. **A digital twin for every plan.** Each plan is tested against simulated reality before approval, and the same twin answers what-if questions.
6. **Two-day-ahead new-start advisor** that balances machines by predicted minutes over a 15-day horizon.
7. **Accessory tracking across departments**, including CT-simulator bookings and QR tags.
8. **Rule Studio.** Plain language becomes versioned, validated configuration, so any department can adopt RAD-SMART without code changes.
9. **Voice-first, multilingual patient loop.** Built on Sarvam AI, with caregivers as first-class recipients and spoken "running late" replies.
10. **Equity by design.** Priorities are visible weights and waits are monitored by patient group.
11. **Meets the clinic where it is.** It works from Excel, paper (OCR) and WhatsApp, with OIS integration added later.
12. **The department's own rules, as configuration.** MHRC bus slots, the 21:00 public-transport limit, accessory reservations, reporting and bladder-filling times, downtime rules and TBI tapering are all data the department can see and change.

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Appendix A. Glossary

Term	Meaning
ABC	Active Breathing Coordinator, a device that helps a patient hold their breath during treatment
CP-SAT	A constraint-programming solver from Google OR-Tools
CT simulator	The CT scanner used to plan radiotherapy; it shares some accessories with the machines
Digital twin	A simulation of the department used to test plans before using them
Fraction	One daily treatment session in a course
Linac	Linear accelerator, the treatment machine (here Versa HD and Halcyon)
MILP	Mixed-integer linear programming, a method that finds the best plan under constraints
MHRC	Institutional respite / hospice accommodation named in the problem statement; its patients travel on the hospital bus
OIS	Oncology information system, such as MOSAIQ or ARIA
P50 / P80	The duration a session will not exceed in 50% / 80% of cases
RTT	Radiation therapy technologist
TBI	Total body irradiation: two long fractions a day for a few days
WTG / WTN	Gross wait (arrival to linac entry) / net wait (appointment to linac entry)

Appendix B. The department's answers (21 September and 24 October 2026)

Dr Akshay Dinesan answered our open questions on behalf of the department, and on 22 September 2026 shared its anonymised arrival and treatment records (Section 2.1). On 24 October 2026 the department described its roles, standing slots, breakdown rules and the screens it needs. Every answer is now part of the PoC's configuration and the CARE application, and the records calibrate its twin.

Question	Department's answer	How RAD-SMART uses it
Complex-case window	10:00–17:00 is hard. 12:00–13:30 is preferred (soft). Protected slots are released if unused	Hard window; soft protected block, open to others when unused
TBI	Two fractions a day of 60–90 min, first of the morning and in the evening; about every 2–3 months, known a month ahead; new starts tapered from a week before	TBI slots blocked; the forecast holds back exactly the new starts that would not fit
Operating hours	Starts 08:30 and may run to about 01:00–02:00; three RTT shifts; ceiling of 90 patients a day; to be reassessed when the second machine arrives	Regular day to 01:00, hard stop 02:00; the test day stays within the ceiling (87 planned plus up to 3 urgent)
Accessories	One breast board and one ABC unit, shared with the CT simulator (11:00–18:00); transfer under 2 min. Linac reservations: both before 11:00; both 16:00–17:00 for new starts and MHRC; ABC 12:00–13:30; breast board linac-only after 18:00; reserved accessories released to the CT simulator when unused	Units, transfers, reservations and CT bookings modelled; bookings checked

Question	Department's answer	How RAD-SMART uses it
Priority slots	Paying patients: preferred time as far as possible. MHRC: protected 16:00–17:00 slots, released if unused; must finish for the 17:00 bus. Public transport: finish before 21:00. Older patients: earlier or convenient slots. Inpatients, dormitory residents and patients living nearby: flexible, can take late-evening slots	Strong preference for paying patients; hard MHRC and 21:00 limits; soft older-patient preference; flexible patients take the late evening
Urgent same-day starts	0–3 a day, arriving in the afternoon; accommodated before 18:00	Two 20-min holds; hard 18:00 limit
Blood irradiation	Daily on the Versa HD, 13:30–14:00	Machine blocked
Second machine	Arrives early 2027; capability matrix pending departmental consensus	Placeholder eligibility in the forecast
Timestamps	Appointment, arrival and treatment-start times are recorded; session durations are not	The records calibrate the twin and give the baseline; the estimate table comes first for durations; room entry and exit captured from day 1 of the pilot
Languages	Mostly Kannada and Tulu; a few English, Hindi and Malayalam	Sarvam for Kannada, Malayalam, Hindi and English; recorded Tulu voice notes with Kannada-script text
Approvals	Live changes and the next-day plan approved by a designated person (senior RTT or oncologist)	Every plan, re-plan and deferral waits for that approval
Roles (24 Oct)	The doctor decides the treatment plan and the machine and approves; the technologist records in and out times, machine status and issues	The course page belongs to the doctor; the day board and breakdown screen to the technologist (Section 6.10)
Standing slots (24 Oct)	A patient keeps one slot until the course completes, unless they ask to change it, and RAD-SMART must accommodate such requests. New starts are done before 17:00; later fractions take evening slots	New-start slot before 17:00 and one standing slot free on every treatment date; slot-change requests proposed and approved (Section 6.4)
Intake (24 Oct)	Hospital number, name, age, diagnosis, total dose, fractions, treatment days (daily, alternate, twice daily, twice weekly, weekly) and a priority checklist with a tick box for every hard and soft rule	The intake form in CARE (Section 6.10)
EMR (24 Oct)	A RADSMART button on the patient's visit page	The button on CARE's visit page; the same pattern applies to the hospital's own EMR
Reporting	At least 20 min before the slot; pelvic patients (cervix, endometrium, rectum, anal canal, bladder) drink 500 mL of water, wait 30 min and report 45 min before	Reporting time and preparation in every message; waits include it
Imaging review	The oncologist may decline to treat; the patient is repositioned and treated at once or later the same day; buffer capacity is needed	Re-queued automatically; holds and the late evening give the buffer
Downtime (24 Oct, replacing the answers of 21 Sep)	Breakdowns last 15 min to 48 h. Up to 60 min: everyone is accommodated with overtime. Over 60 min: patients are sent back according to the expected downtime. New starts are postponed until the machine works. Patients not yet reported are told to postpone travel. A technologist or doctor approves	Built in (Section 6.5)
Data sharing	Anonymised arrival and treatment records for 55 days (October–December 2024) shared on 22 September 2026	Used only as aggregates (medians, percentages, hourly profiles); the spreadsheet stays out of the public repository

Still open:

1. The second machine's capability matrix (techniques, imaging, accessories), pending departmental consensus.
2. The Head of Department's sign-off on quoting aggregate figures from the records in public materials.
3. The department's estimate table: typical minutes per technique, first fraction and accessory.
4. TBI details: the usual course length (we assumed 3 days) and exact slot times (we assumed 08:30–10:00 and 18:30–20:00).
5. How often CT-simulator bookings need each accessory, so the reservations can be tuned.
6. How large the new-start reserve before 17:00 should be (we keep 150 minutes, about five new starts), since it decides how much of the day goes to the evening.

Appendix C. Sample of the optimised plan (synthetic)

Time	Report by	Patient	Technique	Fraction	Planned min	Flags	Today's appointment	Why
08:30	08:10	P060	Breast (breast board)	12/17	8.7	dormitory; paying; breast board	08:30	within 15 min of preferred time 08:30; accessory check: breast board free (CT-simulator bookings and reservations respected); report at 08:10 (20 min before)
08:38	08:18	P067	Palliative 2D/3D	3/4	7.2	older; public transport; paying	08:30	public transport: finish by 21:00; within 15 min of preferred time 08:30; older patient: earlier slot; report at 08:18 (20 min before)
08:46	08:01	P064	IMRT	21/31	8.8	public transport; paying; pelvic	08:15	public transport: finish by 21:00; 31 min later than usual time 08:15; report at 08:01 (45 min before: drink 500 mL of water on arrival)
12:00	11:40	P001	SRT	4/5	27.1	complex; public transport; paying	11:00	complex procedure: senior staff 10:00-17:00; public transport: finish by 21:00; 60 min later than usual time 11:00; report at 11:40 (20 min before)
12:28	12:08	P006	Breast (breast board)	1/19	24.4	new start; public transport; breast board	12:00	new start: finish by 17:00; public transport: finish by 21:00; within 30 min of usual time 12:00; accessory check: breast board free (CT-simulator bookings and reservations respected); report at 12:08 (20 min before)
16:14	15:54	P015	Breast (breast board)	19/21	8.2	older; MHRC; paying; breast board	16:00	MHRC: protected slot 16:00-17:00, back on the 17:00 bus; within 30 min of usual time 16:00; accessory check: breast board free (CT-simulator bookings and reservations respected); report at 15:54 (20 min before)

Time	Report by	Patient	Technique	Fraction	Planned min	Flags	Today's appointment	Why
19:24	18:39	P024	VMAT	14/31	9.7	public transport; pelvic	08:50	public transport: finish by 21:00; 634 min later than usual time 08:50; report at 18:39 (45 min before: drink 500 mL of water on arrival)
21:32	21:12	P048	3D-CRT	17/20	6.7	lives nearby	21:00	32 min later than usual time 21:00; lives nearby: late-evening slot, keeps earlier slots for stricter constraints; report at 21:12 (20 min before)

P024's time moves from 08:50 to 19:24 because this single-day test plans the day from scratch, and today's booking clusters patients into the morning and the late evening. With standing slots (Section 6.4) a patient's time is fixed when the course is approved and does not move from day to day. The full 87-patient plan is in `poc/results/schedule_rad_smart.csv`.

Appendix D. Reproducing the PoC

Requirements: Python 3.10 or newer with numpy, scipy, scikit-learn, pandas and matplotlib (see `poc/requirements.txt`). Patients are generated synthetically. The twin's calibration comes from `poc/data/department_profile.json`, which holds only aggregates of the department's records; `tools/analyse_department_data.py` rebuilds it from the spreadsheet, which is not in the repository.

```
cd poc
pip install -r requirements.txt
python run_poc.py
```

A full run takes about 3 minutes on a laptop and writes `results/results.json`, `results/run_log.txt`, `results/schedule_rad_smart.csv`, `results/web_data.json` and all figures. Random seeds are fixed and ties are broken by patient ID, so a run reproduces the same results, with one exception. When a solve stops at its time limit (here the average-table plan), the best plan found can differ slightly between runs, and so can the numbers that follow from it. Solver timings always vary.